Horizontal resolution dependence of the liquid water path response to droplet number in doubly periodic EAMxx simulations of drizzling stratocumulus

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Abstract. The response of liquid water path (LWP) to cloud droplet number concentration (N_d) in marine stratocumulus results from a competition between precipitation suppression and entrainment, whose representation depends on model grid spacing. We examined it in 288 doubly periodic simulations of the drizzling DYCOMS-II RF02 stratocumulus with the E3SM atmosphere model EAMxx, spanning horizontal grid spacings from 40 km to 312 m, target N_d from 20 to 1000 cm^{-3} and three cloud-droplet sedimentation factors, with the N_d dependence of autoconversion switched on or off. At 10–40 km, where boundary-layer turbulence is entirely parameterised, LWP increased monotonically with N_d, and the susceptibility $d\ln\mathrm{LWP}/d\ln N_\mathrm{d}$ between 100 and 200 cm^{-3} was 0.24–0.27 and vanished without the N_d dependence of autoconversion. With decreasing grid spacing, entrainment generally increased, cloud base rose, rain production at 200 cm^{-3} fell twentyfold, and the LWP maximum moved to about 100 cm^{-3} at 312 m. The weak negative branch above this maximum (susceptibility -0.04 to -0.05) persisted without the N_d dependence of autoconversion and steepened with faster sedimentation, consistent with a sedimentation–entrainment feedback, but it is smaller than one hour of LWP drift in these single, transient runs. A mixed-layer budget associates the thinning at fine grid spacing with entrainment warming and drying that exceed the radiative and surface forcing, but it does not resolve the N_d dependence at sub-kilometre spacing. LWP did not converge, and the turbulence scheme's mixing length was grid-limited at 312 m. In EAMxx, the sign and magnitude of the simulated LWP adjustment therefore depend on grid spacing across the turbulence grey zone.

1 Introduction

Aerosol–cloud interactions remain the largest source of uncertainty in the effective radiative forcing of the climate system (Bellouin et al., 2020; Forster et al., 2021). An increase in cloud condensation nuclei raises the droplet number concentration N_d of a liquid cloud and, at fixed liquid water, its albedo (Twomey, 1974, 1977). The cloud then adjusts, and changes in its liquid water path (LWP) and cloud fraction can amplify or offset this Twomey effect (Stevens and Feingold, 2009; Mülmenstädt and Feingold, 2018; Glassmeier et al., 2021).

In marine stratocumulus, two adjustment pathways of opposite sign dominate. Smaller droplets coalesce less efficiently, which suppresses drizzle and tends to increase LWP (precipitation suppression; Albrecht, 1989). Smaller droplets also sediment more slowly and evaporate more readily near cloud top, which enhances the entrainment of warm, dry free-tropospheric

air and thins the cloud (Wang et al., 2003; Bretherton et al., 2007; Hill et al., 2009; Small et al., 2009). Ackerman et al. (2004) showed with large-eddy simulations (LES) that the sign of the LWP response depends on the balance between the moistening associated with reduced surface precipitation and the drying associated with entrainment, and therefore on the humidity above the inversion. Bretherton et al. (2007) isolated the contribution of droplet sedimentation, which reduces entrainment by removing liquid water from the entrainment zone. Mixed-layer and LES studies relate the competition between these pathways to the time scales of precipitation and entrainment (Wood, 2007; Uchida et al., 2010; Hoffmann et al., 2020, 2024).

Satellite composites of LWP against N_d have an inverted-V shape, with LWP increasing with N_d in clean, precipitating clouds and decreasing in polluted, non-precipitating ones (Chen et al., 2014; Gryspeerdt et al., 2019; Possner et al., 2020). Gryspeerdt et al. (2019) regarded the observed negative slope as an upper bound on the causal effect, because cloud-controlled variations in N_d and meteorological covariability contribute to it. Pollution tracks show LWP responses of both signs that are weak on average and offset about a quarter of the Twomey cooling (Toll et al., 2019), and LES ensembles reproduce the inverted-V shape and indicate long adjustment time scales (Glassmeier et al., 2019, 2021). Present-day spatial co-variability is an imperfect guide to the causal response (Ghan et al., 2016; Arola et al., 2022; Feingold et al., 2022), and in the E3SM global model present-day correlations are insufficient to predict the change in cloud albedo caused by anthropogenic aerosol (Mahfouz et al., 2024). Global climate models (GCMs) generally simulate positive LWP responses (Quaas et al., 2009; Michibata et al., 2016; Gettelman, 2015; Mülmenstädt et al., 2020), although their present-day LWP–N_d correlations can be negative (Mülmenstädt et al., 2024b). Using single-column physics of E3SM with modified sedimentation, Mülmenstädt et al. (2024a) found that the sedimentation–entrainment coupling is active in GCM-type physics and concluded that GCM parameterisations may be able to represent it.

Global storm-resolving models (GSRMs) with grid spacings of a few kilometres are now run routinely (Stevens et al., 2019; Caldwell et al., 2021; Donahue et al., 2024). At these spacings the energy-containing eddies of the stratocumulus-topped boundary layer, a few hundred metres across, are neither resolved nor fully parameterised. This range is known as the turbulence grey zone (Wyngaard, 2004; Honnert et al., 2020), and it poses specific problems for stratocumulus (Boutle et al., 2014; Beare, 2014; Efstathiou and Beare, 2015; de Roode et al., 2019). Even LES of stratocumulus at grid spacings of tens of metres are sensitive to resolution and numerics, mainly through entrainment (Bretherton et al., 1999; Stevens et al., 2005; Pressel et al., 2017; Mellado et al., 2018). In the RF02 single-column model intercomparison of Wyant et al. (2007), LWP after 4–6 h differed by more than a factor of 5 between models, mainly because of differences in turbulence parameterisation. Multiscale models that resolve cloud-scale (Wang et al., 2011) and sub-kilometre (Terai et al., 2020) processes simulated weaker LWP increases with aerosol than conventional configurations, and a global cloud-system-resolving model simulated LWP decreases with aerosol that it attributed to explicitly simulated condensation and evaporation (Sato et al., 2018). Doubly periodic configurations of the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM) show systematic changes in cloud structure between 5 km and a few hundred metres (Bogenschutz et al., 2023a, 2025), and regionally refined global configurations show the importance of vertical resolution for stratocumulus (Bogenschutz et al., 2023b).

To our knowledge, the change in the balance between adjustment pathways across this range of grid spacings has not been quantified within a single model. Such a quantification is needed to interpret aerosol–cloud adjustments in GSRMs and

to use kilometre-scale simulations for constraining parameterisations (Stanford et al., 2026). Here we use a factorial set of doubly periodic simulations of the drizzling DYCOMS-II RF02 stratocumulus with EAMxx, the atmosphere component of E3SM that implements SCREAM. We ask how the LWP susceptibility to N_d depends on horizontal grid spacing between 40 km and 312 m, how much of it can be attributed to precipitation suppression and to droplet sedimentation, which boundary-layer processes accompany its change with grid spacing, and how close the simulations are to the RF02 observations and published intercomparisons. Following Mülmenstädt et al. (2024a), we scale the droplet fall speed; in addition, we switch off the N_d dependence of autoconversion. This mechanism-denial approach allows the two pathways to be separated, subject to the caveats discussed in Sect. 2.5.

2 Model, case and methods

2.1 Model

EAMxx is the C++ atmosphere model of the Energy Exascale Earth System Model (E3SM) and implements SCREAM (Caldwell et al., 2021; Donahue et al., 2024). It solves the nonhydrostatic equations with a spectral-element dynamical core on fourth-order elements (Taylor et al., 2020). Physics is computed on a separate finite-volume grid with 2×2 columns per element ("pg2"; Hannah et al., 2021), so that physics columns are spaced 1.5 times the nominal grid spacing Δx, defined as the mean spacing of the dynamical-core quadrature points. Turbulence, cloud macrophysics and condensation are represented by the Simplified Higher-Order Closure (SHOC; Bogenschutz and Krueger, 2013), an assumed-PDF scheme that diagnoses cloud fraction and liquid water from a prognostic turbulent kinetic energy. SHOC limits its turbulent mixing length to the host grid length, which in this configuration is the nominal Δx. Condensation and evaporation of cloud water are part of SHOC's saturation adjustment and do not depend on droplet size. We verified this in the output, where the vapour–liquid exchange of the microphysics scheme equals its rain evaporation at every level.

Microphysics is computed by the Predicted Particle Properties scheme (P3; Morrison and Milbrandt, 2015). Warm rain forms by the autoconversion and accretion rates of Khairoutdinov and Kogan (2000), without the subgrid cloud-water variance enhancement of autoconversion used in some GCM schemes (Morrison and Gettelman, 2008; Gettelman and Morrison, 2015). Rain fraction is set to one, meaning that rain is assumed to fill the grid cell rather than to overlap with the liquid cloud fraction. Cloud droplets sediment with mass- and number-weighted fall speeds of a gamma size distribution. The P3 sedimentation-tendency diagnostics are not usable in the model version used here, because the tendency buffer is not initialised with the pre-sedimentation mixing ratio; this affects only the diagnostics, not the model state, so the sedimentation flux at cloud top could not be diagnosed. Droplet activation uses a prescribed CCN number concentration from the Simple Prescribed Aerosol option (SPA; Mahfouz et al., 2025). In each layer, P3 raises the in-cloud droplet number to at least the prescribed CCN number multiplied by a scaling factor. Radiation is computed by RRTMGP (Pincus et al., 2019) every third physics step.

We used the doubly periodic configuration of EAMxx (DP-EAMxx, also referred to as DP-SCREAM; Bogenschutz et al., 2023a, 2025), which applies large-scale forcing from an intensive-observation-period (IOP) file to a horizontally periodic domain. All simulations used 128 vertical levels, with a spacing of about 50 m near the inversion. An optional horizontal

("3D") turbulence closure, which applies SHOC's eddy diffusivity in the horizontal, was off in the main experiment and was tested at 625 m and 312 m (Appendix C).

Two runtime options were added to P3 for this study. The first is a sedimentation factor s that multiplies the mass- and number-weighted cloud-droplet fall speeds, analogous to the `cld_sed` parameter of the MG2 scheme (Morrison and Gettelman, 2008; Gettelman and Morrison, 2015) and to the sedimentation experiments of Mülmenstädt et al. (2024a); $s = 1$ is the default. The second makes the prefactor and N_d exponent of the autoconversion rate configurable,

$$\left(\frac{\partial q_\mathrm{r}}{\partial t}\right)_\mathrm{auto} = A\, q_\mathrm{c}^{2.47} \left(\frac{N_\mathrm{d}}{1\,\mathrm{cm}^{-3}}\right)^{-\beta}, \tag{1}$$

where q_c is the in-cloud cloud water mixing ratio ($\mathrm{kg\,kg^{-1}}$) and the rate is in $\mathrm{kg\,kg^{-1}\,s^{-1}}$. The default values $A = 1350$ and $\beta = 1.79$ define the configuration with precipitation suppression on. With suppression off, $\beta = 0$ and $A = 1350 \times 63^{-1.79} = 0.812$, so that the two rates are equal at a diagnosed N_d of $63\,\mathrm{cm^{-3}}$, close to the observed RF02 value of about $55\,\mathrm{cm^{-3}}$ used in the intercomparisons (Ackerman et al., 2009; Wyant et al., 2007). The ratio of the off to the on autoconversion rate at fixed q_c is $(N_\mathrm{d}/63)^{1.79}$, which is 0.13 at $20\,\mathrm{cm^{-3}}$ and about 140 at $1000\,\mathrm{cm^{-3}}$. Switching suppression off therefore removes the direct N_d control of rain formation but also changes the amount of rain substantially, in opposite directions at low and high N_d.

2.2 Case

DYCOMS-II RF02 sampled a nocturnal, heavily drizzling stratocumulus deck over the north-eastern Pacific (Stevens et al., 2003; vanZanten et al., 2005; vanZanten and Stevens, 2005). The case was the basis of an LES intercomparison (Ackerman et al., 2009) and a single-column model intercomparison (Wyant et al., 2007). The observations give a mean LWP of about $100 \pm 20\,\mathrm{g\,m^{-2}}$, an entrainment rate of 6.0–7.6 $\mathrm{mm\,s^{-1}}$ (Faloona et al., 2005), a mean surface drizzle rate of $0.35\,\mathrm{mm\,d^{-1}}$ and an evaporated fraction of about 73 % of the cloud-base precipitation (vanZanten et al., 2005; Wyant et al., 2007). We used the E3SM single-column IOP file for this case. The initial boundary layer has $\theta_l = 288.3$ K and $q_t = 9.45\,\mathrm{g\,kg^{-1}}$ below an inversion at 795 m, which gives a lifting condensation level of about 400 m. Surface sensible and latent heat fluxes were prescribed as 16 and 93 $\mathrm{W\,m^{-2}}$, large-scale subsidence was applied with a divergence of $3.75 \times 10^{-6}\,\mathrm{s^{-1}}$, and temperature and moisture were nudged towards the initial profiles above 650 hPa with a time scale of 1 h. Shortwave radiation was off.

The configuration departs from the intercomparison protocol in three respects. Winds were not nudged and there was no Coriolis force, so the initial wind shear decayed freely. The forcing was held constant for 12 h, twice the length of the protocol, to examine the evolution of the adjustment. The surface fluxes were prescribed, which removes the feedback of boundary-layer warming and drying on the fluxes; we estimate this feedback in Sect. 4.4. Unless stated otherwise we analyse hours 10–12, and Sect. 3.1 uses hours 4–6 for comparison with the intercomparisons.

2.3 Experiment design

Table 1 lists the factors. The main matrix crosses eight grid spacings, six target droplet numbers, three sedimentation factors and precipitation suppression on or off, which gives 288 simulations with 3D turbulence off. A further 72 simulations repeated

Table 1. Experiment factors. Every combination was run for 12 h with 3D turbulence off (288 runs). The 625 m and 312 m grids were repeated with 3D turbulence on (72 runs, of which 71 completed).

Factor	Levels	Implementation
Grid spacing Δx	40, 20, 10, 5, 2.5, 1.25 km; 625, 312 m	Table 2
Target N_{d}	20, 50, 100, 200, 500, 1000 cm^{-3}	SPA CCN scaling factor (exponent 1)
Sedimentation factor s	0.5, 1, 2	multiplies cloud-droplet fall speeds in P3
Precipitation suppression	on, off	Eq. (1) with $(A, \beta) = (1350, 1.79)$ or $(0.812, 0)$
3D turbulence	off; on at 625 and 312 m	SHOC eddy diffusivity applied horizontally

the 625 m and 312 m grids with 3D turbulence on; one of them, at 312 m, target $N_{\mathrm{d}} = 1000\,\mathrm{cm}^{-3}$, $s = 2$ with suppression on, did not complete and is excluded, so 71 enter Appendix C.

Table 2 gives the numerical settings. The domain is 240 km wide for $\Delta x \leq 10$ km, and 600 and 300 km at 40 and 20 km so that the coarse domains still contain 10×10 physics columns. The physics time step decreases from 100 s at $\Delta x \geq 5$ km to 15 s at 312 m, and the radiation interval correspondingly from 300 to 45 s. The ratio of physics to dynamics time step is 12 for $\Delta x \geq 2.5$ km and 24 for $\Delta x \leq 1.25$ km, chosen for stability of the dynamical core. Time step and radiation interval therefore co-vary with Δx, and their effects cannot be separated from those of grid spacing in this experiment. The 625 m and 312 m runs were integrated in restart segments, which we verified to be bit-for-bit identical to continuous integrations.

The in-cloud N_{d} produced by a given CCN scaling factor depends on the height of the cloud, because the prescribed CCN number decreases with height. For a target of 200 cm^{-3}, the activation floor decreases from about 345 cm^{-3} below 300 m to about 150 cm^{-3} near 900 m (Appendix B, Fig. B1b). At $\Delta x \geq 10$ km, where the cloud extends down to about 200 m, the diagnosed in-cloud N_{d} is 21–64 % above target, partly because small SHOC cloud fractions near cloud base inflate the in-cloud value. At 312 m, where the cloud lies between about 650 and 1000 m, it is between 21 % below and 11 % above target. The diagnosed in-cloud N_{d}, defined as the median over cloudy grid cells (cloud water above $10^{-5}\,\mathrm{kg\,kg}^{-1}$) of the 3D snapshots at 10 and 12 h, changes by less than 17 % between 6 and 12 h at $\Delta x \leq 5$ km and by up to 25 % at $\Delta x \geq 10$ km.

2.4 Diagnostics and notation

Diagnostics were computed from domain-mean profiles averaged over 15 min, from instantaneous column fields every 15 min, from 2-hourly 3D snapshots, and from cloud-top fields every 15 min. The cloud-top fields were produced by an output operator that follows the AeroCom convention of reporting the liquid cloud-top fraction and the fraction-weighted droplet number and effective radius of the uppermost liquid layer. LWP is the domain mean. Cloud cover is reported both as the fraction of columns with LWP above $10\,\mathrm{g\,m}^{-2}$, and as the maximum-overlap cloud cover computed from SHOC's liquid cloud fraction in the 3D snapshots, which includes subgrid cloud. The inversion height z_{i} is the height at which the domain-mean liquid potential temperature θ_l crosses the midpoint between its boundary-layer (below 400 m) and free-tropospheric (1400 m) values. Cloud base z_{b} is the lowest level at which the domain-mean liquid cloud fraction reaches half its maximum; it is quantised by the 25–

Table 2. Numerical settings. Δx is the nominal (dynamics) grid spacing; physics columns are $1.5\Delta x$ apart. N_{col} is the number of physics columns, Δt_{phys} the physics time step, Δt_{dyn} the dynamics time step and ν_{top} the sponge-layer diffusivity.

Δx	domain (km)	elements	N_{col}	Δt_{phys} (s)	Δt_{dyn} (s)	ν_{top} ($\mathrm{m^2\,s^{-1}}$)
40 km	600	5×5	100	100	8.33	1.2×10^5
20 km	300	5×5	100	100	8.33	6.0×10^4
10 km	240	8×8	256	100	8.33	3.0×10^4
5 km	240	16×16	1024	100	8.33	1.5×10^4
2.5 km	240	32×32	4096	75	6.25	7500
1.25 km	240	64×64	16 384	60	2.50	1667
625 m	240	128×128	65 536	30	1.25	833
312 m	240	256×256	262 144	15	0.625	417

All runs used 128 levels, radiation every third physics step and one macrophysics–microphysics subcycle per physics step.

50 m level spacing. The entrainment rate is $w_{\mathrm{e}} = dz_{\mathrm{i}}/dt - w_{\mathrm{s}}(z_{\mathrm{i}})$, where w_{s} is the IOP subsidence velocity (negative downward) at the mean z_{i} and dz_{i}/dt is a linear fit over hours 8–12. The 4 h fit is less sensitive than a 2 h fit to the 50 m level spacing at the inversion, across which z_{i} rises by about 70 m at 40 km and about 200 m at 312 m over 12 h. Vertical-velocity variance and skewness of the resolved flow were computed on the physics grid from the 3D snapshots at 10 and 12 h, with $w = -\omega/(\rho g)$, and averaged over the boundary layer. They are compared with SHOC's subgrid variance, which SHOC diagnoses from its turbulent kinetic energy and is therefore not an independent second moment. Decoupling is measured as the difference in q_t between the lowest and highest quarters of the boundary layer (Jones et al., 2011). The spectral peak wavelength λ of LWP is the maximum of the radially averaged power spectrum (Wood and Hartmann, 2006); with a 240 km domain it is quantised at $240/n$ km.

We define the LWP susceptibility between two otherwise identical runs at target droplet numbers a and b as

$$S_{a\to b} = \frac{\ln \mathrm{LWP}_b - \ln \mathrm{LWP}_a}{\ln N_{\mathrm{d},b} - \ln N_{\mathrm{d},a}}, \tag{2}$$

where LWP is averaged over hours 10–12 and N_{d} is the diagnosed in-cloud value. The subscripts refer to the target N_{d}, while the slope uses the diagnosed N_{d}. We report three branches, $S_{20\to50}$, $S_{100\to200}$ and $S_{200\to1000}$, and denote susceptibilities with precipitation suppression on and off by S_{on} and S_{off}. Because N_{d} is diagnosed from snapshots while LWP varies continuously, we recomputed S for each of the eight 15 min intervals in hours 10–12 with fixed N_{d} and report the standard deviation. These 15 min samples are autocorrelated and come from a single realisation, so this spread measures temporal sampling only and is not an estimate of the uncertainty that an initial-condition ensemble would provide. Comparisons across grid spacings at fixed N_{d} use LWP and other quantities interpolated linearly in $\ln N_{\mathrm{d}}$ to fixed diagnosed values, because the diagnosed N_{d} at a given target differs between grid spacings.

2.5 Factorial decomposition

For each grid spacing and branch, we fitted $S_{\rm on}(s) = a_{\rm on} + b_{\rm on} s$ and $S_{\rm off}(s) = a_{\rm off} + b_{\rm off} s$ over $s = 0.5$, 1 and 2, and decomposed the susceptibility at $s = 1$ as

$$S_{\rm on}(1) = (a_{\rm on} - a_{\rm off}) + b_{\rm off} + (b_{\rm on} - b_{\rm off}) + a_{\rm off}. \tag{3}$$

The first term is the precipitation-suppression contribution extrapolated to zero sedimentation, the second the sedimentation contribution in the absence of precipitation suppression, the third the interaction between sedimentation and precipitation suppression, and the fourth a residual that includes radiative effects of the effective radius and any non-linearity in s. The sum reproduces $S_{\rm on}(1)$ to within the fit error. Three caveats apply. The extrapolation to $s = 0$ uses three points with one residual degree of freedom, and $S_{\rm off}$ is clearly convex in s on the clean branch at sub-kilometre spacing, so $a_{\rm on}$ and $a_{\rm off}$ are uncertain there. On and off runs at the same target have slightly different diagnosed $N_{\rm d}$, so their slopes span different $\ln N_{\rm d}$ intervals. Most importantly, the off configuration is matched at $63\,{\rm cm^{-3}}$ and therefore precipitates heavily at high $N_{\rm d}$; at 40 km and target $1000\,{\rm cm^{-3}}$, surface precipitation is $0.83\,{\rm mm\,d^{-1}}$ with suppression off and $0.01\,{\rm mm\,d^{-1}}$ with suppression on. The off runs at coarse spacing thus measure the sensitivity to sedimentation in a strongly precipitating state, whereas at 312 m the off runs precipitate only 0.01–0.17 $\rm mm\,d^{-1}$. The partition between the first two terms at coarse spacing therefore depends on the choice of the matching $N_{\rm d}$ and should be read as specific to this design.

2.6 Mixed-layer budget

To relate the LWP response to boundary-layer processes, we evaluated a bulk budget of the domain-mean boundary layer for every run in 2 h windows, following mixed-layer theory (Lilly, 1968; Stevens, 2002) and the LWP-tendency framework of van der Dussen et al. (2014). For a well-mixed layer of depth $z_{\rm i}$ with conserved variables θ_l and q_t,

$$\frac{d\theta_l}{dt} = \frac{1}{z_{\rm i}}\left[\frac{H}{\rho c_p} + w_{\rm e}\,\Delta\theta_l - \frac{F_{\rm rad}}{\rho c_p} + \frac{P}{\rho c_p}\right], \tag{4}$$

$$\frac{dq_t}{dt} = \frac{1}{z_{\rm i}}\left[\frac{E}{\rho L_v} + w_{\rm e}\,\Delta q_t - \frac{P}{\rho L_v}\right], \tag{5}$$

$$\frac{dz_{\rm i}}{dt} = w_{\rm e} + w_{\rm s}(z_{\rm i}), \tag{6}$$

where H and E are the surface sensible and latent heat fluxes, $F_{\rm rad}$ is the radiative cooling of the layer below $z_{\rm i} + 150$ m, P is the surface rain rate in energy units, and $\Delta\theta_l$ and Δq_t are the differences between the values 100 m above $z_{\rm i}$ and the mixed-layer means over 0.1–0.8 $z_{\rm i}$. Cloud base was taken as the lifting condensation level of mixed-layer air, so that $dz_{\rm b}/dt = (\partial z_{\rm b}/\partial\theta_l)\,d\theta_l/dt + (\partial z_{\rm b}/\partial q_t)\,dq_t/dt$, with sensitivities of 120–130 $\rm m\,K^{-1}$ and -190 to -210 $\rm m\,(g\,kg^{-1})^{-1}$ computed with the saturation formula of Bolton (1980). Assuming LWP $\propto h^2$ with cloud thickness $h = z_{\rm i} - z_{\rm b}$ and fixed adiabatic fraction and cloud cover,

$$\frac{d\ln{\rm LWP}}{dt} = \frac{2}{h}\left(\frac{dz_{\rm i}}{dt} - \frac{dz_{\rm b}}{dt}\right), \tag{7}$$

which separates into contributions from entrainment (through w_e in dz_i/dt and in Eqs. 4 and 5), subsidence, radiation, surface fluxes and precipitation.

Because w_e was diagnosed from the simulated dz_i/dt, the budget's dz_i/dt equals the simulated value by construction, and only the cloud-base tendency and the LWP–thickness relation are predicted. The budget should therefore be read as a diagnostic decomposition. It also assumes a well-mixed layer, whereas the low-N_d boundary layers at $\Delta x \leq 5$ km are decoupled (Sect. 3.8) and the sub-kilometre runs have cloud cover well below one. Pooled over all 1440 run-windows with 3D turbulence off, the predicted and simulated tendencies are correlated at 0.89 for θ_l, 0.87 for q_t and 0.88 for LWP. The skill varies strongly with grid spacing (Appendix A, Table A1). For LWP it is high at $\Delta x \geq 5$ km (correlation 0.79–0.90) and low at sub-kilometre spacing (0.07 at 625 m and 0.44 at 312 m, with errors comparable to the signal). For the differences between N_d levels in hours 10–12, which determine S, the correlation is 0.85–0.93 at 40, 20 and 5 km and -0.02 to 0.57 at $\Delta x \leq 2.5$ km.

3 Results

3.1 Comparison with RF02 observations and intercomparisons

Table 3 compares hours 4–6 of the runs at target $N_d = 50\,\mathrm{cm^{-3}}$ (diagnosed 42–67 $\mathrm{cm^{-3}}$) and $s = 1$ with the observations and with the intercomparisons, which averaged over the last two of their 6 h. The runs at $\Delta x \geq 5$ km have an LWP of 156–159 $\mathrm{g\,m^{-2}}$, above the observed $100 \pm 20\,\mathrm{g\,m^{-2}}$ and similar to the DHARMA LES value of 144 $\mathrm{g\,m^{-2}}$ reported by Wyant et al. (2007). Their entrainment rate of 4.1 mm $\mathrm{s^{-1}}$ is below the observed 6.0–7.6 mm $\mathrm{s^{-1}}$ and close to the LES value of 4.5 mm $\mathrm{s^{-1}}$, and their surface precipitation of 0.12–0.14 mm $\mathrm{d^{-1}}$ is below the observed 0.35 mm $\mathrm{d^{-1}}$. With decreasing grid spacing, LWP decreases to 98 $\mathrm{g\,m^{-2}}$ at 312 m and the entrainment rate increases to 7.7 mm $\mathrm{s^{-1}}$, at the upper end of the observed range. Cloud base rises from about 300 m to 580 m, compared with the lifting condensation level of about 400 m of the initial mixed layer. The 625 m runs are closest to the observations overall, with an LWP of 119 $\mathrm{g\,m^{-2}}$, an entrainment rate of 6.3 mm $\mathrm{s^{-1}}$, surface precipitation of 0.28 mm $\mathrm{d^{-1}}$ and an evaporated fraction of 0.73, equal to the observed value. No single grid spacing matches all observed quantities, and LWP, entrainment and precipitation at all spacings lie within the spread of the single-column models, whose LWP ranged from 30 to 162 $\mathrm{g\,m^{-2}}$ and entrainment rate from 1.5 to 6.8 mm $\mathrm{s^{-1}}$ (Wyant et al., 2007). Ackerman et al. (2009) found that their LES ensemble with drizzle and sedimentation had a steady LWP consistent with the measurements and an entrainment rate at the low end of the measured range, and that it underestimated precipitation. The coarse EAMxx runs share the low entrainment and low precipitation, while the sub-kilometre runs entrain more than the LES. We did not run an LES of the same configuration, so a converged reference with this model physics is not available.

3.2 Cloud structure and mean state across grid spacings

Figure 1 shows LWP at hour 12 for $s = 1$ with precipitation suppression on. As the grid is refined, the LWP spectral peak shortens from about 30 km at 2.5 km to 13–22 km at 625 and 312 m, the domain-mean LWP decreases, and clear gaps appear. Interpolated to a diagnosed in-cloud N_d of 200 $\mathrm{cm^{-3}}$, LWP is 273 $\mathrm{g\,m^{-2}}$ at 40 km, 204 $\mathrm{g\,m^{-2}}$ at 5 km and 73 $\mathrm{g\,m^{-2}}$ at 312 m,

Table 3. Hours 4–6 at target $N_\mathrm{d} = 50\,\mathrm{cm}^{-3}$ and sedimentation factor 1, compared with RF02 observations and published simulations. P_sfc is surface precipitation and f_evap the fraction of rain production that evaporates before reaching the surface. Observations: LWP and f_evap as compiled by Wyant et al. (2007), w_e from Faloona et al. (2005), P_sfc from vanZanten et al. (2005). LES and SCM values are averages over hours 4–6 from Table 2 of Wyant et al. (2007) (standard vertical resolution).

Δx	supp.	LWP ($\mathrm{g\,m^{-2}}$)	w_e ($\mathrm{mm\,s^{-1}}$)	P_sfc ($\mathrm{mm\,d^{-1}}$)	f_evap	z_b (m)	z_i at 6 h (m)
40 km	on	159	4.2	0.14	0.85	303	849
20 km	on	158	4.1	0.13	0.85	303	849
10 km	on	157	4.1	0.12	0.86	299	850
5 km	on	156	4.1	0.13	0.85	303	851
2.5 km	on	132	4.5	0.68	0.52	432	864
1.25 km	on	132	5.1	0.52	0.61	443	870
625 m	on	119	6.3	0.28	0.73	506	888
312 m	on	98	7.7	0.10	0.84	578	913
40 km	off	171	4.2	0.05	0.91	303	851
20 km	off	171	4.2	0.05	0.91	303	852
10 km	off	170	4.2	0.05	0.92	313	852
5 km	off	168	4.2	0.06	0.90	313	853
2.5 km	off	141	4.5	0.52	0.55	432	866
1.25 km	off	142	5.3	0.42	0.63	443	872
625 m	off	125	6.8	0.23	0.73	501	891
312 m	off	103	7.7	0.08	0.85	572	914
Observations		100 ± 20	6.0–7.6	0.35	0.73		
DHARMA LES		144	4.5				
SCMs (9 models)		30–162	1.5–6.8				

and the maximum-overlap cloud cover decreases from 1.00 at $\Delta x \geq 2.5$ km to 0.92 at 312 m. At target $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$ the sub-kilometre clouds break into small cells of high LWP, and the maximum-overlap cloud cover at 312 m is 0.68 (0.50 with the LWP threshold). The 5 km fields contain stripe-like structures with a wavelength of about six physics columns, close to the effective resolution of the dynamical core, which suggests that the circulations at this spacing are partly numerical. Between 40 km and 312 m at target $200\,\mathrm{cm}^{-3}$, the inversion rises from 863 to 1016 m, cloud base rises from 195 to 720 m, the cloud layer thins from 668 to 296 m, and the mixed layer becomes 1.8 K warmer. The 40 and 20 km runs are nearly indistinguishable. The 10 km runs differ from them at high N_d (LWP 312 instead of $342\,\mathrm{g\,m^{-2}}$ at target $1000\,\mathrm{cm}^{-3}$), where their column LWP standard deviation reaches up to $27\,\mathrm{g\,m^{-2}}$ across configurations, compared with at most $5\,\mathrm{g\,m^{-2}}$ at 20 and 40 km. Resolved motions carry no measurable vertical-velocity variance at $\Delta x \geq 10$ km (Sect. 3.8), so these runs approximate a single-column configuration of the model physics, with the 10 km runs beginning to develop weak horizontal structure.

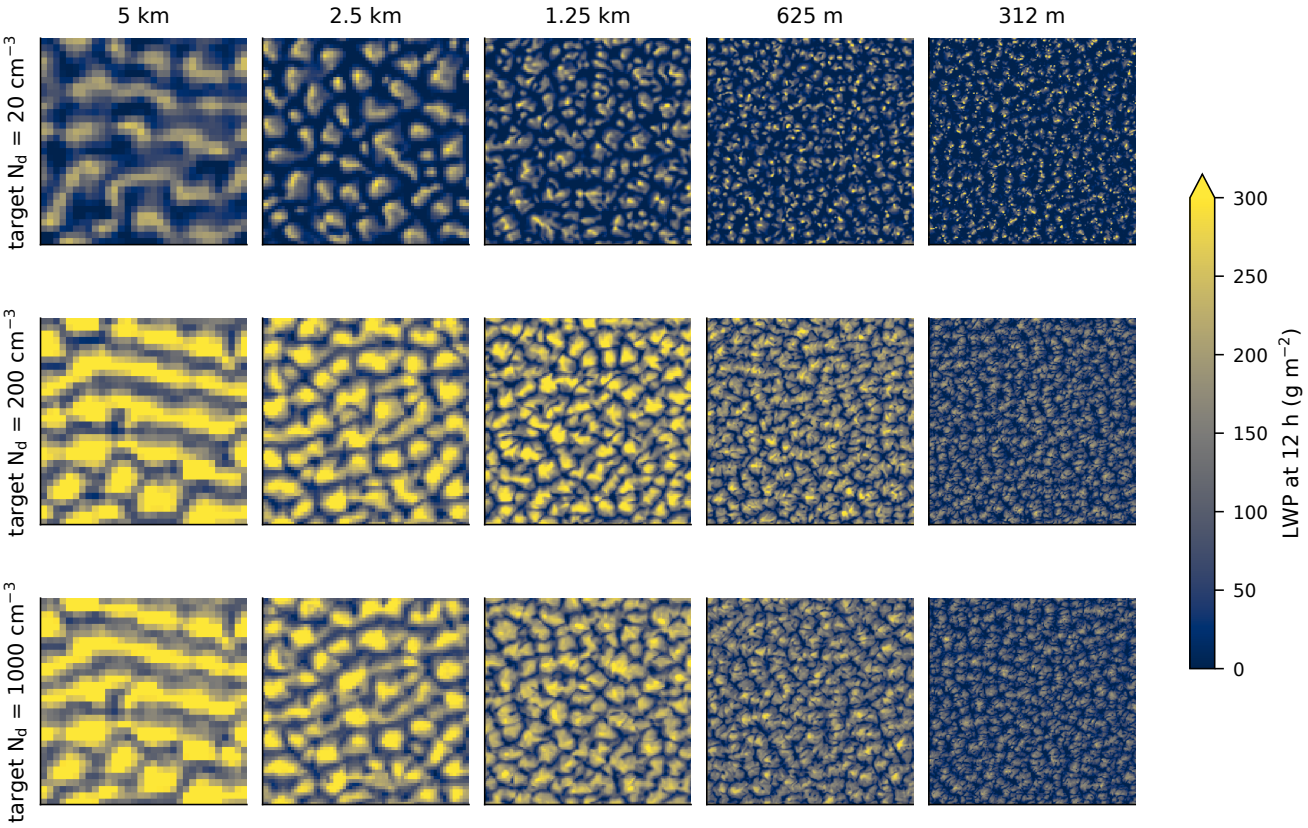


Figure 1. Instantaneous LWP at hour 12 over the 240 km domain for target N_d of 20, 200 and 1000 cm^{-3} (rows) and grid spacings from 5 km to 312 m (columns), for sedimentation factor 1 with precipitation suppression on and 3D turbulence off.

3.3 LWP response to droplet number

Figure 2 and Table 4 show the LWP response to N_d. With precipitation suppression on, LWP increases monotonically with N_d at $\Delta x \geq 10$ km, from 135 to 342 $\mathrm{g\,m}^{-2}$ at 40 km. With decreasing grid spacing, the curve flattens at high N_d and then turns over. The LWP maximum moves from target $N_\mathrm{d} \geq 1000\,\mathrm{cm}^{-3}$ at $\Delta x \geq 10$ km to 500 cm^{-3} at 5 km, 200 cm^{-3} from 2.5 km to 625 m, and 100 cm^{-3} (diagnosed 84 cm^{-3}) at 312 m. The resulting shape resembles the inverted-V shape in satellite composites and LES ensembles (Gryspeerdt et al., 2019; Glassmeier et al., 2021), but in these simulations it appears only at $\Delta x \leq 5$ km, and the negative slopes are an order of magnitude smaller than the observed ones, which Gryspeerdt et al. (2019) regarded as an upper bound on the causal effect.

With precipitation suppression off (Fig. 2b), LWP decreases slightly with N_d at $\Delta x \geq 10$ km (from 234 to 208 $\mathrm{g\,m}^{-2}$ at 40 km) and at 312 m (from 79 to 64 $\mathrm{g\,m}^{-2}$). At 5 km it increases monotonically, from 135 to 160 $\mathrm{g\,m}^{-2}$, and at 2.5 km to 625 m it increases between target 20 and 50 cm^{-3} ($S_{20\to50} = 0.08$–0.20) and decreases slightly at higher N_d. Surface precipitation

Table 4. Domain-mean LWP ($\mathrm{g\,m^{-2}}$, hours 10–12) for sedimentation factor 1 and 3D turbulence off, at the target N_d ($\mathrm{cm^{-3}}$) of each column. Bold marks the maximum with precipitation suppression on. Diagnosed in-cloud N_d at a given target differs between grid spacings (Sect. 2.3).

Δx	supp.	20	50	100	200	500	1000
40 km	on	135	188	246	295	333	**342**
	off	234	221	216	211	209	208
20 km	on	135	187	244	292	328	**336**
	off	233	220	215	211	209	207
10 km	on	134	186	240	281	308	**312**
	off	232	217	213	208	206	204
5 km	on	82	137	184	214	**222**	218
	off	135	149	153	158	159	160
2.5 km	on	67	119	163	**177**	175	170
	off	117	137	139	138	137	136
1.25 km	on	65	111	149	**157**	154	149
	off	116	127	127	124	122	122
625 m	on	52	98	120	**122**	119	115
	off	103	109	108	103	101	99
312 m	on	40	72	**76**	74	72	70
	off	79	79	72	69	66	64

(Fig. 2d) decreases steeply with N_d when suppression is on, and more steeply at finer spacing; at 312 m it is $0.011\,\mathrm{mm\,d^{-1}}$ at target $100\,\mathrm{cm^{-3}}$ and at most $0.002\,\mathrm{mm\,d^{-1}}$ at higher N_d. The entrainment rate (Fig. 2c) is 3.1–4.5 $\mathrm{mm\,s^{-1}}$ at $\Delta x \geq 10$ km and increases with N_d at $\Delta x \leq 5$ km whether suppression is on or off. It is not monotonic in Δx: at target $N_\mathrm{d} \geq 100\,\mathrm{cm^{-3}}$ it is larger at 5 km (7.0–8.1 $\mathrm{mm\,s^{-1}}$) than at 2.5 km (5.7–7.3 $\mathrm{mm\,s^{-1}}$), and it increases again towards 312 m (8.0–8.3 $\mathrm{mm\,s^{-1}}$).

Figure 3 and Table 5 give the corresponding susceptibilities. With suppression on and $s = 1$, $S_{100\to200}$ decreases monotonically as Δx decreases, from 0.27 at 40 km to 0.20 at 5 km, 0.11 at 2.5 km, 0.07 at 1.25 km, 0.03 at 625 m and -0.05 at 312 m. The polluted branch $S_{200\to1000}$ changes sign between 5 km (0.01) and 2.5 km (-0.03) and lies between -0.03 and -0.04 at finer spacing. The clean branch $S_{20\to50}$ increases from 0.39 at $\Delta x \geq 10$ km to 0.65–0.76 at $\Delta x \leq 2.5$ km. At sub-kilometre spacing, part of this increase is an increase in cloud cover; at 312 m the cloud-cover part of $S_{20\to50}$ is 0.33 with the maximum-overlap definition (0.53 with the LWP threshold). On the high-N_d branches cloud cover contributes less than 0.03. The temporal-sampling standard deviation is at most 0.03 at $\Delta x \leq 2.5$ km and up to 0.1 at $\Delta x \geq 5$ km, where LWP oscillates on time scales of about 15 min.

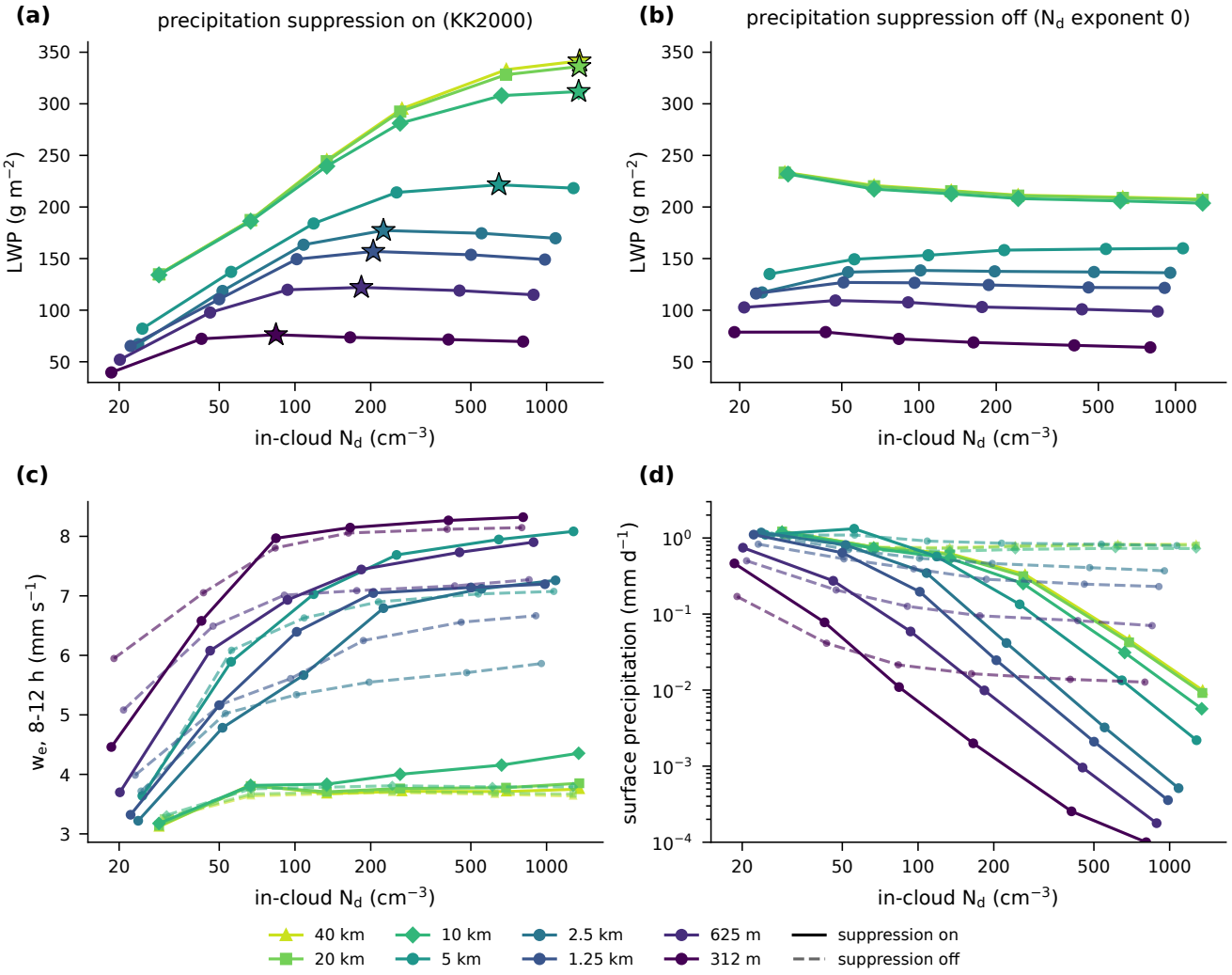


Figure 2. Response to N_d at every grid spacing (colours and symbols) for sedimentation factor 1. (a) LWP with precipitation suppression on; stars mark the maximum. (b) LWP with suppression off. (c) Entrainment rate (8–12 h fit) and (d) surface precipitation, with suppression on (solid) and off (dashed); values below 10^{-4} mm d^{-1} are plotted at the lower axis limit. Values are hour 10–12 means against the diagnosed in-cloud N_d.

At 312 m, the negative $S_{100\to200}$ corresponds to an LWP difference of 2.6 g m^{-2} (3.4 %) between targets 100 and 200 cm^{-3}, and $S_{200\to1000}$ to a difference of 4.1 g m^{-2} between 200 and 1000 cm^{-3}. These differences are smaller than the LWP decrease of each run in one hour (Sect. 3.4), and initial-condition spread in domain-mean LWP of stratocumulus simulations is typically of a few per cent. Without an ensemble, the sign of these small susceptibilities at 312 m is therefore not established. The dependence on s is more systematic. At $\Delta x \le 2.5$ km with suppression on, $S_{200\to1000}$ is -0.01 to -0.02 for $s = 0.5$ and

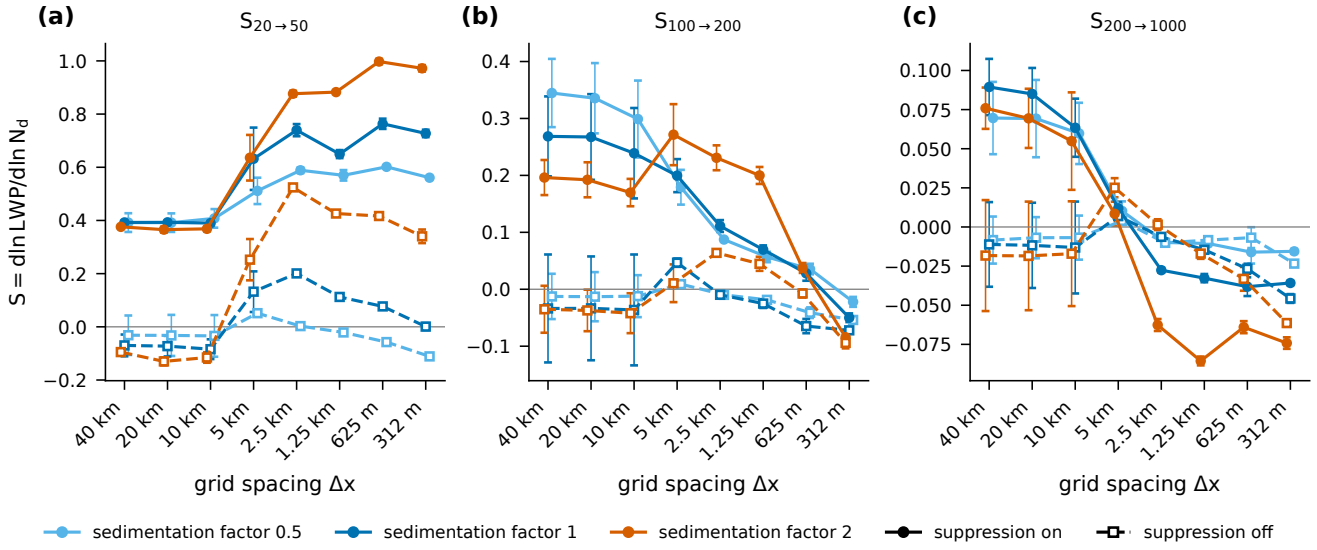


Figure 3. LWP susceptibility on the (a) clean, (b) moderate and (c) polluted branches against grid spacing (logarithmic axis). Colours: sedimentation factor. Solid lines and filled circles: precipitation suppression on; dashed lines and open squares: off. Bars show the standard deviation of S over the eight 15 min intervals in hours 10–12, a measure of temporal sampling only. Symbols are offset horizontally for clarity.

-0.06 to -0.09 for $s = 2$ (Fig. 3c). On the moderate branch, $S_{100\to200}$ increases with s at 2.5 and 1.25 km (from 0.09 to 0.23 and from 0.06 to 0.20) and decreases with s at 312 m (from -0.02 to -0.08). With suppression off, S is within ±0.13 of zero at $\Delta x \geq 10$ km on all branches. At $\Delta x \leq 1.25$ km and $s \leq 1$ it is negative on the two high-N_d branches (-0.07 and -0.05 at 312 m for $s = 1$).

3.4 Temporal evolution of the susceptibility

The susceptibilities evolve throughout the 12 h (Fig. 4). At $\Delta x \geq 10$ km, $S_{100\to200}$ increases from 0.12 in hours 2–4 to 0.24–0.27 in hours 10–12, as the high-N_d clouds continue to gain the water that low-N_d clouds lose to precipitation (Fig. 4d); at 40 km and high N_d, LWP is still increasing by up to 15 $\mathrm{g\,m^{-2}\,h^{-1}}$ in hours 10–12. At 1.25 km and 625 m, $S_{100\to200}$ also increases slowly (from 0.00 to 0.07 and 0.03), whereas at 312 m it decreases from 0.00 to -0.05. The polluted branch $S_{200\to1000}$ becomes more negative with time at all $\Delta x \leq 2.5$ km, from about -0.01 in hours 2–4 at $\Delta x \leq 1.25$ km and from 0.04 at 2.5 km to between -0.03 and -0.04 in hours 10–12. The clean branch is already large in hours 2–4 at 625 and 312 m (0.56–0.63) and at 1.25 km (0.51) and increases slowly. At $\Delta x \leq 2.5$ km, LWP decreases in 98 % of the runs during hours 10–12, by up to 4.4 $\mathrm{g\,m^{-2}\,h^{-1}}$ or 6 % $\mathrm{h^{-1}}$, and none of the susceptibilities at sub-kilometre spacing had levelled off at 12 h. The adjustment time scales in these runs are therefore longer than the simulation, as in LES of non-precipitating stratocumulus, where LWP continues to adjust over a day or more (Glassmeier et al., 2021).

Table 5. Susceptibility $S = d\ln \mathrm{LWP}/d\ln N_\mathrm{d}$ for sedimentation factor 1 and 3D turbulence off. In parentheses: standard deviation over the eight 15 min intervals in hours 10–12. The last column is the target N_d (cm^{-3}) of maximum LWP with suppression on.

	$S_{20\to50}$		$S_{100\to200}$		$S_{200\to1000}$		target N_d
Δx	on	off	on	off	on	off	at max LWP
40 km	0.39 (0.01)	−0.07 (0.04)	0.27 (0.07)	−0.03 (0.09)	0.09 (0.02)	−0.01 (0.03)	1000
20 km	0.39 (0.01)	−0.07 (0.04)	0.27 (0.08)	−0.03 (0.09)	0.08 (0.02)	−0.01 (0.03)	1000
10 km	0.39 (0.01)	−0.08 (0.04)	0.24 (0.08)	−0.04 (0.10)	0.06 (0.02)	−0.01 (0.03)	1000
5 km	0.63 (0.12)	0.13 (0.08)	0.20 (0.03)	0.05 (0.01)	0.01 (<0.01)	0.01 (<0.01)	500
2.5 km	0.74 (0.02)	0.20 (0.01)	0.11 (0.01)	−0.01 (<0.01)	−0.03 (<0.01)	−0.01 (<0.01)	200
1.25 km	0.65 (0.02)	0.11 (<0.01)	0.07 (0.01)	−0.03 (0.01)	−0.03 (<0.01)	−0.01 (<0.01)	200
625 m	0.76 (0.02)	0.08 (<0.01)	0.03 (0.01)	−0.06 (0.01)	−0.04 (0.01)	−0.03 (<0.01)	200
312 m	0.73 (0.01)	0.00 (0.01)	−0.05 (0.01)	−0.07 (0.01)	−0.04 (<0.01)	−0.05 (<0.01)	100

3.5 Precipitation suppression, sedimentation and their interaction

Figure 5 shows the decomposition of Eq. (3) at $s = 1$. At $\Delta x \geq 10$ km, the precipitation-suppression term at zero sedimentation is 0.34–0.39 on the moderate branch and 0.07–0.08 on the polluted branch, and the sedimentation term derived from the off runs is small (-0.02 to -0.01). The interaction term on the moderate branch is -0.07 to -0.08, because $S_{100\to200}$ with suppression on decreases strongly with s (from 0.35 to 0.20 between $s = 0.5$ and 2 at 40 km), whereas the off runs barely respond. This sensitivity to sedimentation at coarse spacing is larger than at 312 m ($dS_\mathrm{on}/ds = -0.10$ at 40 km and -0.04 at 312 m). It is unlikely to act through entrainment, which changes by less than 0.6 mm s^{-1} between $s = 0.5$ and 2 at $\Delta x \geq 10$ km (Fig. 7a). Deposition of cloud droplets at the surface is also unlikely to explain it, because the estimated deposition at the lowest model level exceeds 0.02 mm d^{-1} only at $\Delta x \geq 10$ km, target 20 cm^{-3}, $s = 2$ with suppression off (up to 0.25 mm d^{-1}, or about 20 % of surface precipitation). The sensitivity to sedimentation at coarse spacing therefore depends on the autoconversion formulation, presumably through the vertical redistribution of cloud water relative to where rain forms, but we have not isolated its mechanism. In addition, because of the pivot dependence described in Sect. 2.5, the statement that precipitation suppression accounts for the positive response at coarse spacing holds for this design and should not be read as a pivot-independent attribution.

At finer spacing, the precipitation-suppression term on the moderate branch decreases to 0.12 at 5 km, 0.05–0.10 between 2.5 km and 625 m and 0.04 at 312 m, and on the polluted branch it is 0.00–0.02 at $\Delta x \leq 5$ km. On the polluted branch at 2.5 and 1.25 km, the negative $S_{200\to1000}$ comes mainly from the interaction term (-0.04 and -0.05) and the residual (-0.01), while the sedimentation term derived from the off runs is near zero ($+0.01$ and -0.01). At 625 m and 312 m the sedimentation term (-0.016 and -0.024), the interaction (-0.015 at both) and the residual (-0.004 and -0.016) all contribute. On this branch, a larger sedimentation factor increases LWP more at target 200 than at 1000 cm^{-3}, so $S_{200\to1000}$ becomes more negative with

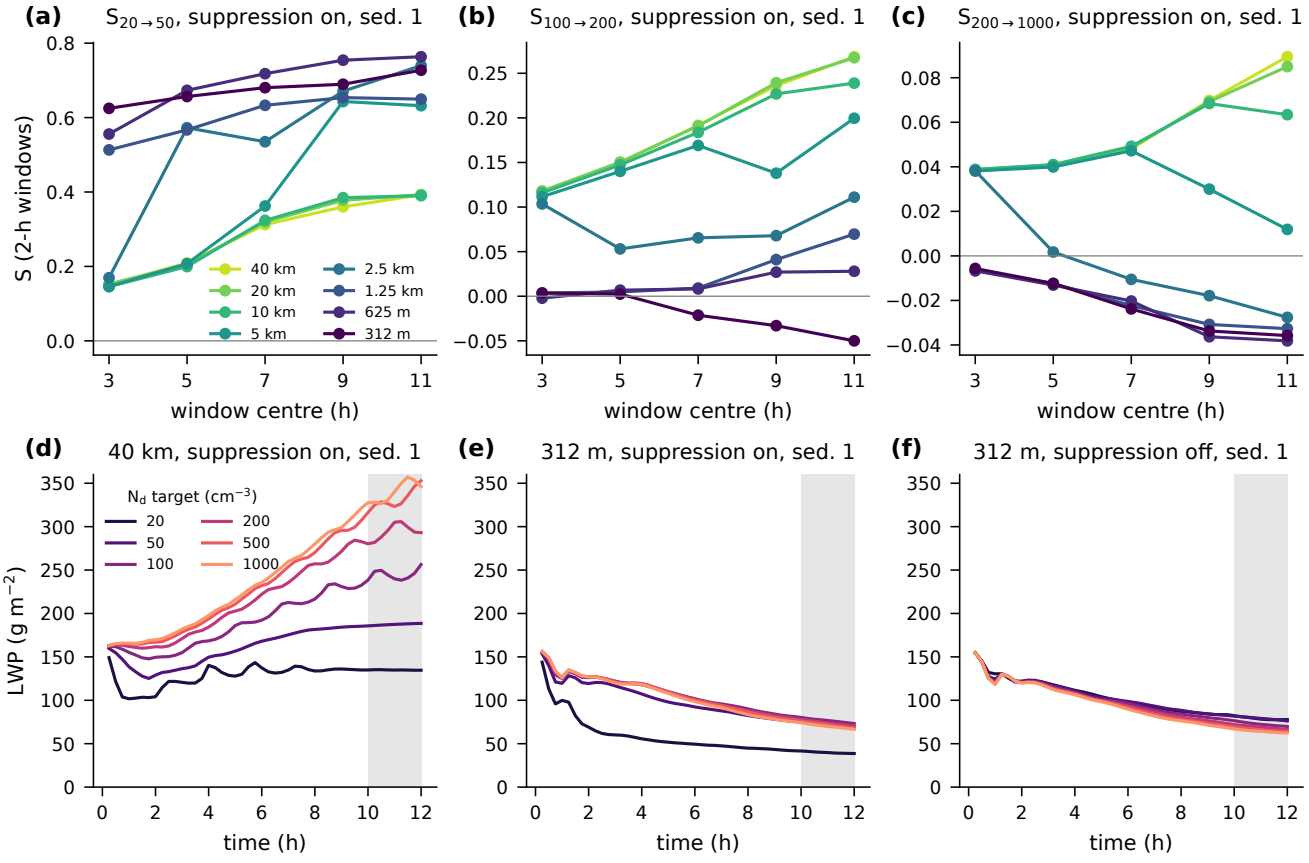


Figure 4. (a–c) Susceptibility on the three branches for consecutive 2 h windows (suppression on, sedimentation factor 1), coloured by grid spacing. (d–f) LWP time series for each target N_d at 40 km with suppression on and at 312 m with suppression on and off. Grey shading marks the analysis window.

s, consistent with the mechanism of Bretherton et al. (2007). At 2.5 and 1.25 km the moderate branch behaves differently, with positive sedimentation and interaction terms (both about 0.05), so that $S_{100\to200}$ increases with s.

On the clean branch at $\Delta x \le 5$ km, the sedimentation term is positive (0.13 at 5 km and 0.30–0.34 at $\Delta x \le 2.5$ km). At target $N_\mathrm{d} = 20\,\mathrm{cm}^{-3}$, cloud-top effective radii are 12–14 μm for $s = 1$ with suppression on, and a larger sedimentation factor lowers LWP most at the lowest N_d. At these droplet sizes, sedimentation mainly removes cloud water, as drizzle does, and its effect on entrainment is secondary. The clean-branch residual is large and negative at $\Delta x \le 2.5$ km (-0.16 to -0.28), but it is the least well constrained term because S_off is convex in s (-0.11, 0.00 and 0.34 for $s = 0.5$, 1 and 2 at 312 m). We interpret it only as an indication that pathways other than a linear sedimentation effect, possibly including radiative effective-radius effects on thin, broken clouds, contribute on this branch.

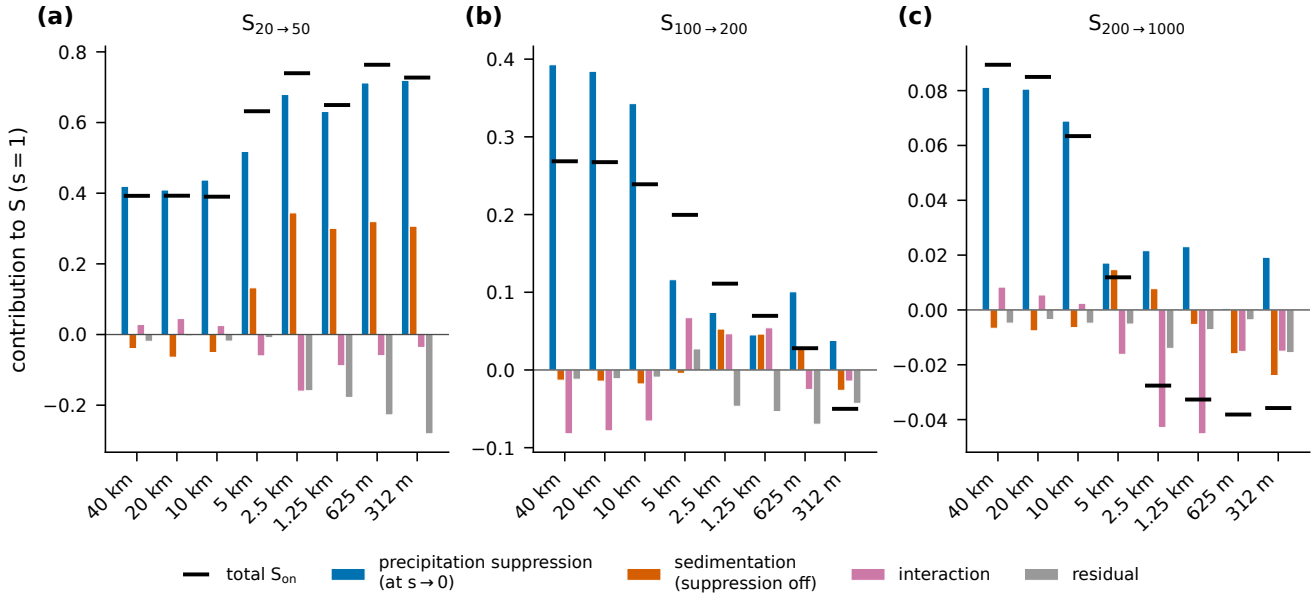


Figure 5. Decomposition (Eq. 3) of the susceptibility with suppression on at sedimentation factor 1 (black bars) on the (a) clean, (b) moderate and (c) polluted branches. Blue: precipitation suppression extrapolated to zero sedimentation; orange: sedimentation without precipitation suppression; pink: interaction; grey: residual. The partition at coarse spacing depends on the matching N_d of the suppression-off configuration (Sect. 2.5).

3.6 Resolution dependence of the precipitation sink

The effect of precipitation suppression on LWP is limited by the size of the precipitation sink, and Fig. 6 shows that this sink decreases with grid spacing. Rain production (autoconversion plus accretion) at target 200 cm^{-3} decreases from 0.55 mm d^{-1} at 40 km to 0.03 mm d^{-1} at 312 m (Fig. 6a), as the cloud thins from about 650 to about 300 m (Fig. 6d) and the KK2000 rates decrease steeply with cloud water. At target 20 cm^{-3} the decrease is smaller, from about 2.0 to 1.2 mm d^{-1}. As cloud base rises, the fraction of rain production that evaporates below cloud increases, at target 200 cm^{-3} from 0.40 at 40 km to 0.93 at 312 m (Fig. 6b), consistent with the importance of subcloud evaporation in drizzling stratocumulus (Wood, 2005; Wyant et al., 2007). At the same time the entrainment drying flux $-\rho L_v w_\mathrm{e} \Delta q_t$ roughly doubles, from 67 to 128 W m^{-2} at target 200 cm^{-3} (Fig. 7c). The ratio of surface precipitation to entrainment drying (Fig. 6c), which is the diagnostic that Ackerman et al. (2004) used to explain the sign of the LWP response, therefore decreases with grid spacing by a factor of about 4 at target 20 cm^{-3} (from 0.65 to 0.18) and by a factor of about 300 at targets 200–1000 cm^{-3}. The raining fraction of the domain (surface rain above 0.5 mm d^{-1}) is below 0.2 % at $\Delta x \leq 1.25$ km for target $N_\mathrm{d} \geq 200\,\mathrm{cm}^{-3}$. When surface precipitation is negligible, precipitation suppression has little effect on LWP, and the decrease of LWP with N_d associated with entrainment

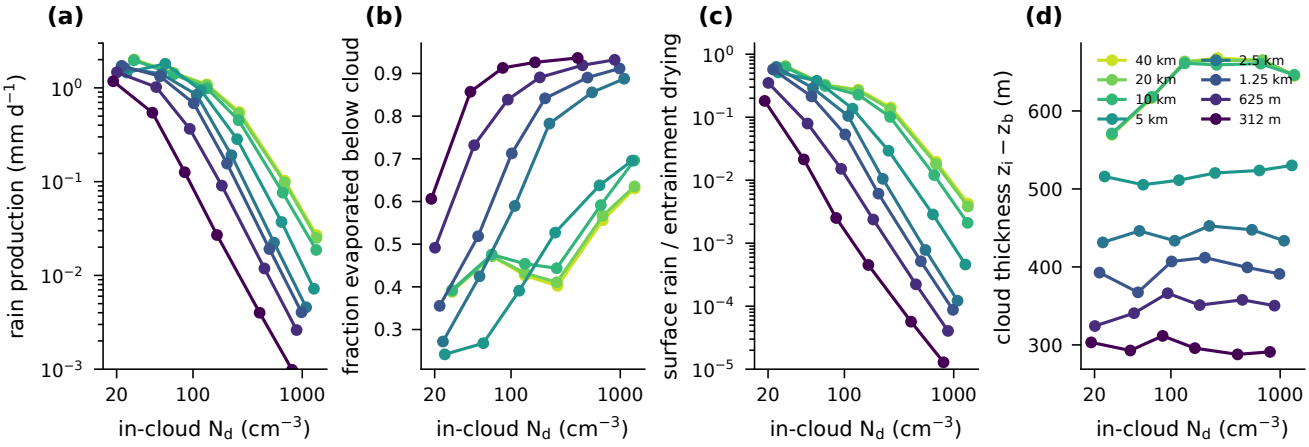


Figure 6. Precipitation budget against in-cloud N_d at every grid spacing (suppression on, sedimentation factor 1). (a) Column rain production (autoconversion plus accretion). (b) Fraction of the production that evaporates before reaching the surface. (c) Ratio of surface precipitation to the entrainment drying flux. (d) Cloud thickness $z_\mathrm{i} - z_\mathrm{b}$.

dominates (Ackerman et al., 2004; Wood, 2007; Hoffmann et al., 2020). In these simulations, the N_d at which this transition occurs depends on grid spacing.

3.7 Entrainment and the boundary-layer budget

At target 200 cm^{-3} with suppression on, the entrainment rate is 3.7 mm s^{-1} at 40 km, 7.7 mm s^{-1} at 5 km, 6.8 mm s^{-1} at 2.5 km, 7.4 mm s^{-1} at 625 m and 8.1 mm s^{-1} at 312 m. The boundary-layer radiative cooling is 80–88 W m^{-2} in all these runs, so the change in w_e mostly reflects a change in entrainment efficiency. We define the ratio of entrainment warming to radiative cooling as

$$\eta = \frac{\rho c_p w_\mathrm{e} \Delta\theta_l}{F_\mathrm{rad}}. \tag{8}$$

At $\Delta x \geq 10$ km, η is 0.45–0.67 (Fig. 7b). At $\Delta x \leq 5$ km it exceeds one for target N_d of 100–200 cm^{-3} and above, with the largest values at 5 km (1.07–1.25 for $N_\mathrm{d} \geq 100$ cm^{-3}) and at 312 m (1.2–1.3). When η exceeds one, entrainment warming exceeds radiative cooling, and with the surface sensible heat flux the mixed layer warms by 0.07–0.10 K h^{-1} at 312 m. Entrainment drying at 312 m (105–131 W m^{-2} for $N_\mathrm{d} \geq 50$ cm^{-3}) likewise exceeds the prescribed latent heat flux of 93 W m^{-2}, whereas at $\Delta x \geq 10$ km it is 55–78 W m^{-2} (Fig. 7c). The response of w_e to the sedimentation factor at fixed N_d (Fig. 7a) is largest at $\Delta x \leq 5$ km and low N_d, where doubling the fall speed relative to $s = 0.5$ reduces w_e by 1.1–2.4 mm s^{-1} at target $N_\mathrm{d} \leq 50$ cm^{-3}; at target $N_\mathrm{d} \geq 500$ cm^{-3} the reduction is 0.2–0.8 mm s^{-1}. At $\Delta x \geq 10$ km the reduction is less than 0.6 mm s^{-1} at all N_d. This is the expected sign of the sedimentation–entrainment feedback (Bretherton et al., 2007; Ackerman et al., 2009), and its larger magnitude at fine spacing indicates that the coupling between sedimentation and entrainment is stronger where part of the boundary-layer circulation is resolved.

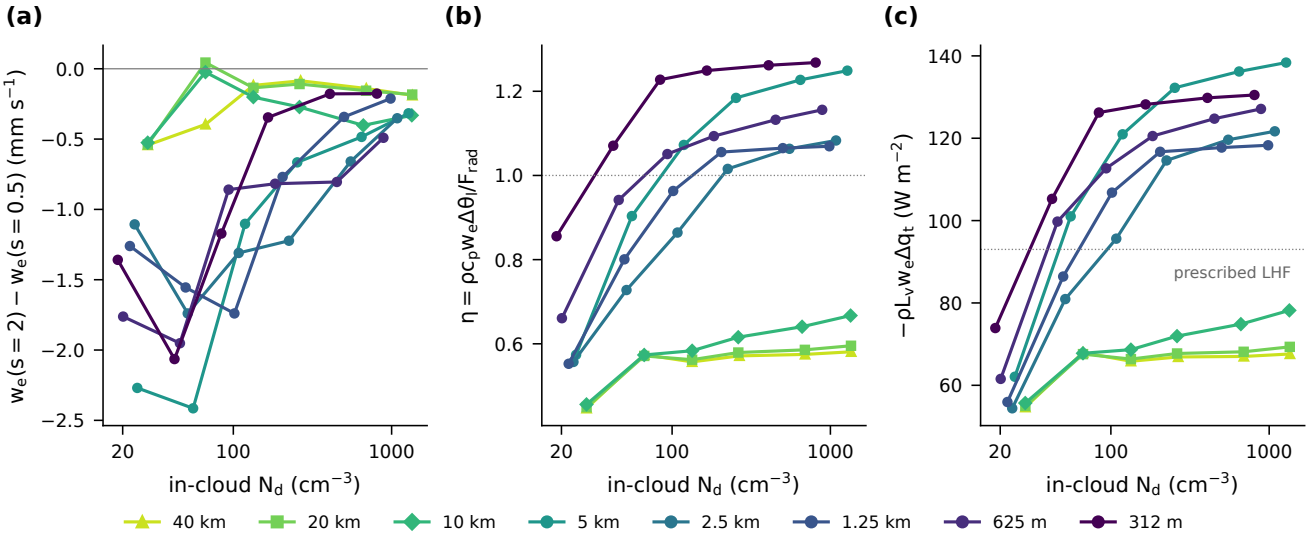


Figure 7. (a) Change in the entrainment rate between sedimentation factors 2 and 0.5 at fixed target N_d (suppression on). (b) Ratio η of entrainment warming to boundary-layer radiative cooling and (c) entrainment drying flux, with the prescribed surface latent heat flux (dotted line), for suppression on and sedimentation factor 1.

The mixed-layer budget (Sect. 2.6) expresses these fluxes as contributions to the LWP tendency (Fig. 8). In every run, radiative cooling and surface fluxes thicken the cloud, and entrainment and subsidence thin it. At 40 km, where the budget reproduces both the tendencies and their differences between N_d levels (Table A1), the entrainment contribution is about $-9\,\%\,h^{-1}$ at all N_d, and the precipitation contribution decreases in magnitude from $-10\,\%\,h^{-1}$ at target $20\,cm^{-3}$ to nearly zero at $N_d \geq 500\,cm^{-3}$. The increase of LWP with N_d at 40 km is therefore associated with the removal of the precipitation sink, consistent with Sect. 3.5. At 312 m, the entrainment contribution is about three times larger (-27 to $-32\,\%\,h^{-1}$ for $N_d \geq 50\,cm^{-3}$), and it is partly offset by larger radiative (21–24 $\%\,h^{-1}$) and surface-flux (12–13 $\%\,h^{-1}$) contributions; the precipitation contribution is below $1\,\%\,h^{-1}$ in magnitude for $N_d \geq 100\,cm^{-3}$. The budget sum is negative at all N_d (-2 to $-5\,\%\,h^{-1}$), consistent in sign and magnitude with the simulated thinning (-4 to $-6\,\%\,h^{-1}$). At 625 m the budget sum at high N_d (-5.4 to $-5.7\,\%\,h^{-1}$) is more negative than the simulated trend (-1.9 to $-2.9\,\%\,h^{-1}$). The budget therefore suggests that the general thinning at fine grid spacing is associated with the increase of entrainment relative to the radiative and surface forcing. At $\Delta x \leq 2.5$ km, however, the net N_d dependence of the budget sum (about $2\,\%\,h^{-1}$ at 312 m) is no larger than the budget error, and the entrainment term itself varies non-monotonically with N_d (-29, -29, -27, -31 and $-32\,\%\,h^{-1}$ for targets 50 to $1000\,cm^{-3}$ at 312 m). The budget does not establish which process sets the small negative susceptibilities at sub-kilometre spacing.

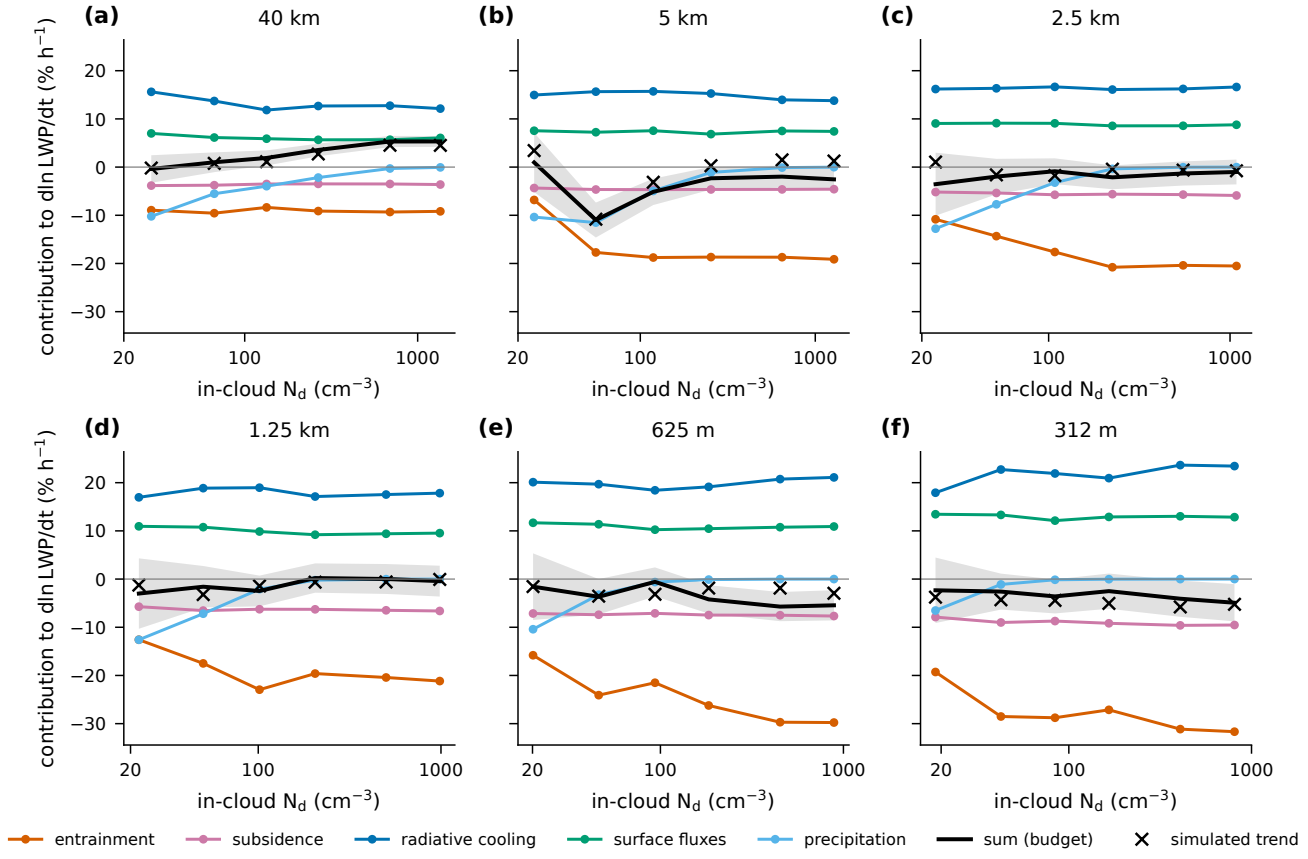


Figure 8. Mixed-layer LWP budget (Eq. 7) for hours 10–12, expressed as contributions to $d\ln\mathrm{LWP}/dt$, at (a) 40 km, (b) 5 km, (c) 2.5 km, (d) 1.25 km, (e) 625 m and (f) 312 m (suppression on, sedimentation factor 1). The black line is the budget sum, with shading of $\pm$ the root-mean-square error of the budget at that grid spacing (Table A1); crosses are the simulated LWP trend. The entrainment rate in the budget is diagnosed over each 2 h window, whereas Figs. 2 and 7 use the 8–12 h fit.

3.8 Subgrid and resolved turbulence

At $\Delta x \geq 10$ km, all vertical-velocity variance is parameterised (Fig. 9a), and SHOC's boundary-layer variance lies between 0.28 and 0.41 m^2 s^{-2}. At 5 and 2.5 km this variance is 0.07–0.11 m^2 s^{-2}, and it decreases to 0.03 m^2 s^{-2} at 312 m, while the resolved variance increases. The resolved share of the total is 3–7 % at 5 km, 4–9 % at 2.5 km, 9–19 % at 1.25 km, 28–38 % at 625 m and 54–67 % at 312 m, so that a third of the variance remains subgrid even at 312 m. Part of this repartition is imposed by the closure. SHOC's mixing length is capped at the nominal grid spacing, and at 312 m the domain-mean mixing length is within 5 % of this cap at 90 % of the boundary-layer levels, compared with 25 % at 625 m and none at $\Delta x \geq 1.25$ km (Appendix B, Fig. B1a). At 312 m, SHOC's eddy diffusivity is therefore directly limited by the grid spacing throughout the boundary layer. The large change between 10 and 5 km, where the resolved share is still small, is also not explained by resolved turbulence.

At 5 km SHOC's boundary-layer mean mixing length is about two thirds of its value at 10 km (Fig. B1a), even though the cap is not active, which points to a change in the SHOC solution itself, possibly coupled to the stripe-like circulations noted in Sect. 3.2. We have not diagnosed this transition further.

The resolved vertical-velocity skewness (Fig. 9b), which is meaningful only at $\Delta x \leq 5\,\mathrm{km}$, is positive at low N_d (up to 1.9 at 312 m for target $20\,\mathrm{cm}^{-3}$), indicating narrow, strong updrafts in a drizzle-decoupled boundary layer. At high N_d it is -0.7 to -1.0 at 2.5 and 1.25 km and -0.3 to -0.7 at 625 m, the sign expected for radiatively driven downdrafts in a well-mixed stratocumulus layer (Wood, 2012); the transition occurs at $N_\mathrm{d} \approx 50$–$100\,\mathrm{cm}^{-3}$. At 312 m the skewness stays positive at all N_d (0.09–0.24 for target $N_\mathrm{d} \geq 100\,\mathrm{cm}^{-3}$). With a third of the variance still subgrid and the mixing length at its cap, this may reflect near-grid-scale updrafts. Drizzle decouples the boundary layer at low N_d (Fig. 9c). At $\Delta x \leq 5\,\mathrm{km}$, Δq_t is 1.1–1.6 $\mathrm{g\,kg}^{-1}$ at target $20\,\mathrm{cm}^{-3}$ and 0.21–0.64 $\mathrm{g\,kg}^{-1}$ at $1000\,\mathrm{cm}^{-3}$, consistent with below-cloud evaporation stabilising the subcloud layer (Stevens et al., 1998; Bretherton and Wyant, 1997; Jones et al., 2011). In the single-column limit, q_t decreases upward by 1.2–1.7 $\mathrm{g\,kg}^{-1}$ at all N_d across a deep cloud layer.

The LWP spectral peak changes less between 625 and 312 m than LWP itself (Fig. 9d; Appendix D). At 5 and 2.5 km the peak wavelengths of 48 and 30 km correspond to 6.4 and 8 physics-column spacings and are probably limited by the effective resolution of the dynamical core. At 625 and 312 m the peak lies at 13–22 km, an aspect ratio λ/z_i of 13–20. Observed closed mesoscale cells have aspect ratios of up to about 30–40 for boundary layers about 1 km deep (Wood and Hartmann, 2006), and LES with large domains also produce cells that are wider than these (Kazil et al., 2017; Zhou et al., 2018; Zhou and Bretherton, 2019). The sub-kilometre cells are therefore on the narrow side of the observed range, and the 2.5 km value (33) agrees better with the observations, although it is probably resolution-limited.

Switching on 3D turbulence changes LWP by less than 1.5 $\mathrm{g\,m}^{-2}$ ($< 1.5\,\%$), cloud cover by less than 0.007 and w_e by -0.13 to $+0.02\,\mathrm{mm\,s}^{-1}$ at 625 m and 312 m (Appendix C). The 3D turbulence option therefore has a negligible effect on the results, which is expected because SHOC's horizontal diffusion is small compared with the horizontal dissipation of the dynamical core.

3.9 Convergence

None of the bulk quantities converges with grid spacing (Fig. 10). At fixed diagnosed N_d, LWP decreases by 10–23 % from 1.25 km to 625 m and by a further 21–40 % from 625 to 312 m, the larger decreases occurring at higher N_d. The maximum-overlap cloud cover and the cloud thickness also continue to decrease. For LWP at target $N_\mathrm{d} \geq 50\,\mathrm{cm}^{-3}$, the change over the last halving of the grid spacing exceeds that over the previous one by a factor of 1.3–2.0, so the sequence is not in an asymptotic range and an extrapolation to zero grid spacing is not meaningful. The entrainment rate is non-monotonic, with a minimum at 2.5–1.25 km, as expected in the grey zone where parameterised and resolved contributions trade places (Honnert et al., 2020; Boutle et al., 2014). The clean-branch susceptibility (0.65–0.76 at $\Delta x \leq 2.5\,\mathrm{km}$) varies less with grid spacing than the bulk quantities, but it varies with s from 0.56 to 0.97 at 312 m. The vertical grid (about 50 m at the inversion, compared with about 5 m in the RF02 LES; Ackerman et al., 2009) was held fixed, and vertical resolution at the inversion is known to

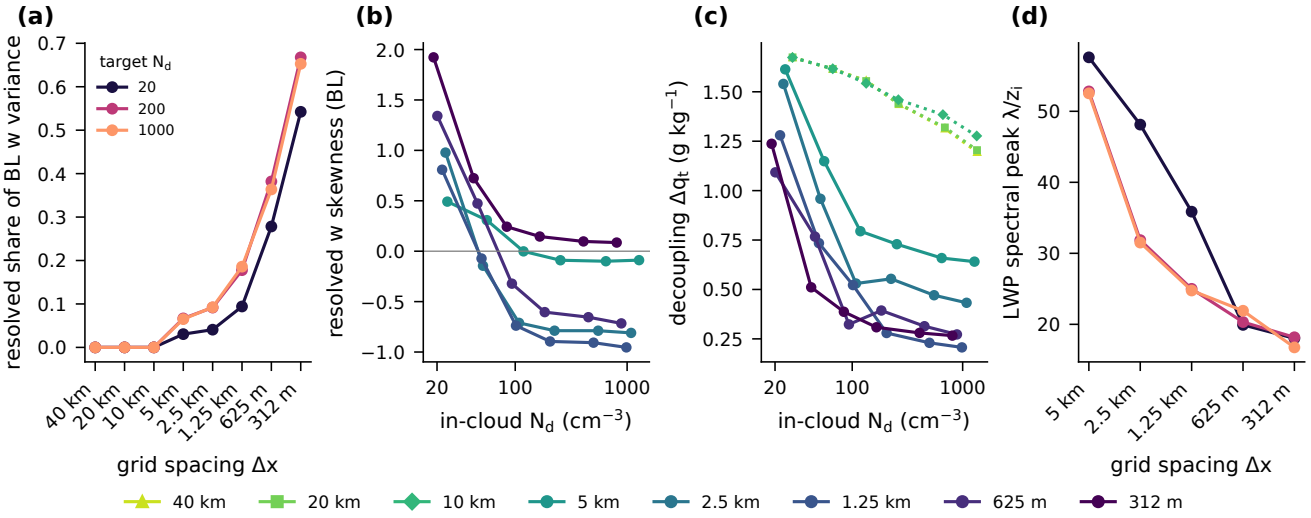


Figure 9. Boundary-layer structure (suppression on, sedimentation factor 1). (a) Resolved share of boundary-layer vertical-velocity variance against grid spacing for three target N_d. (b) Resolved vertical-velocity skewness over the boundary layer ($\Delta x \leq 5\,\mathrm{km}$). (c) Decoupling Δq_t against in-cloud N_d (dotted lines: $\Delta x \geq 10\,\mathrm{km}$). (d) LWP spectral peak wavelength relative to z_i ($\Delta x \leq 5\,\mathrm{km}$).

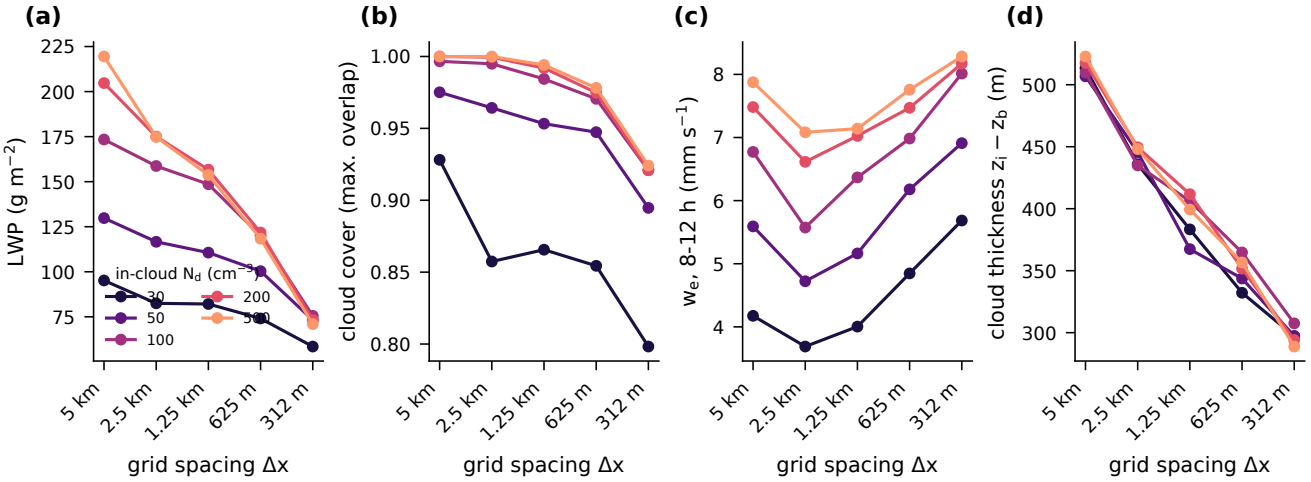


Figure 10. Hour 10–12 means against grid spacing ($\Delta x \leq 5\,\mathrm{km}$) at fixed diagnosed in-cloud N_d, interpolated in $\ln N_\mathrm{d}$ (suppression on, sedimentation factor 1). (a) LWP, (b) maximum-overlap cloud cover including SHOC's subgrid cloud fraction, (c) entrainment rate (8–12 h fit) and (d) cloud thickness.

control entrainment in LES and in SCREAM (Stevens et al., 2005; Bogenschutz et al., 2023b). The sign of $S_{100\to200}$ at 312 m may change with further horizontal or vertical refinement.

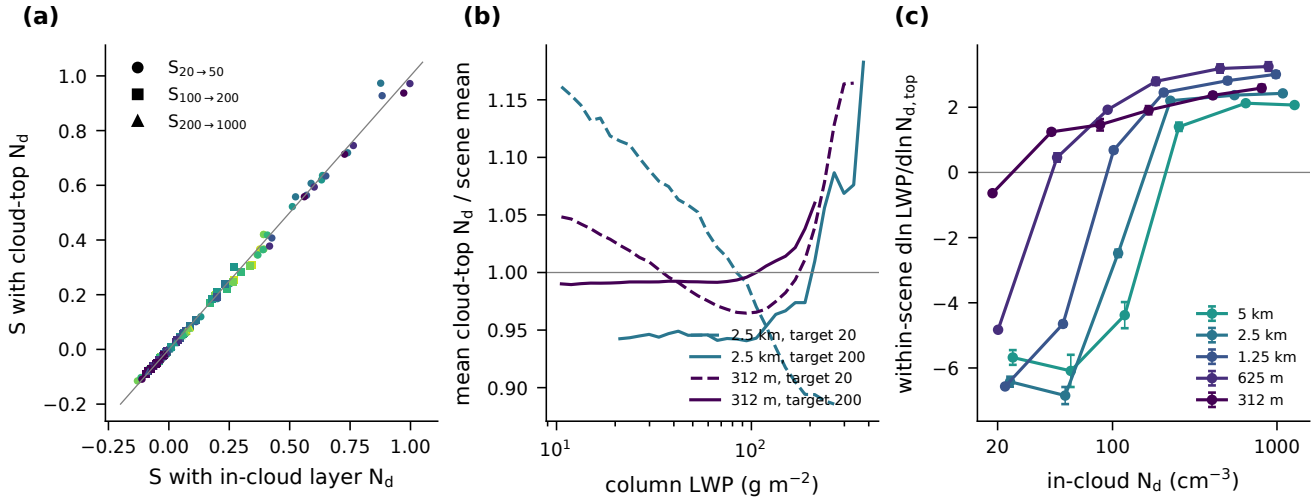


Figure 11. (a) Susceptibility computed with model-diagnosed cloud-top N_d against that computed with in-cloud layer N_d for all branches and configurations (3D turbulence off), coloured by grid spacing. (b) Mean cloud-top N_d relative to the scene mean as a function of column LWP, at 2.5 km and 312 m for target N_d of 20 and 200 cm^{-3} (hours 10–12). (c) Within-scene regression slope of column $\ln$LWP on column cloud-top $\ln N_\mathrm{d}$, with standard errors over the eight instants (suppression on, sedimentation factor 1, $\Delta x \leq 5$ km).

3.10 Cloud-top droplet number and within-scene covariability

Satellite estimates of LWP adjustments use N_d retrieved from cloud-top effective radius and optical depth under adiabatic assumptions (Grosvenor et al., 2018). The model-diagnosed cloud-top N_d used here is not such a retrieval. It is 44–72 % of the in-cloud layer value at $\Delta x \geq 10$ km and 77–95 % at 312 m, and because this ratio varies little with N_d at a given grid spacing, susceptibilities computed with cloud-top N_d differ from those computed with layer N_d by at most 0.10, 0.04 and 0.01 on the three branches (Fig. 11a). This agreement follows largely from the near-constant ratio and says nothing about retrieval errors.

Within each domain CCN are horizontally uniform, so the spatial covariation of LWP and cloud-top N_d arises from cloud processes. We regressed column $\ln$LWP on column cloud-top $\ln N_\mathrm{d}$ over cloudy columns at each 15 min instant in hours 10–12, restricting the analysis to $\Delta x \leq 5$ km because the columns at coarser spacing are nearly identical. The slopes range from -7.9 to $+3.4$, with standard errors over instants of at most 0.8 and correlations between -0.86 and $+0.56$ (Fig. 11c). At 2.5–5 km and low N_d they are negative, because collision–coalescence depletes N_d in the thickest, drizzling columns (Fig. 11b). At 625 and 312 m and target $N_\mathrm{d} \geq 50\,\mathrm{cm}^{-3}$ they are positive (0.4–3.3), because cloud-top N_d is reduced in thin, evaporating cloud edges. At 312 m and target 200 cm^{-3}, for example, the within-scene slope is $+1.9$, whereas $S_{100\to200} = -0.05$. That cloud-controlled N_d variability produces spatial LWP–N_d relationships unrelated to the causal response is well established (Gryspeerdt et al., 2019; Arola et al., 2022; Mülmenstädt et al., 2024b; Mahfouz et al., 2024). The additional result here is that the sign of this covariation changes with grid spacing, because the relative importance of drizzle depletion in thick cloud and evaporation in thin cloud edges depends on how much of the cloud field is resolved.

4 Discussion

4.1 Resolution dependence of the dominant adjustment pathway

At $\Delta x \geq 10\,\mathrm{km}$, where the simulations approximate single-column physics, entrainment in EAMxx is nearly insensitive to droplet sedimentation, the clouds are thick enough to precipitate substantially, and LWP increases monotonically with N_d. This behaviour corresponds to the precipitation-suppression mechanism of Albrecht (1989). At finer grid spacing, entrainment increases ($\eta > 1$), and the boundary layer warms, dries and deepens. The cloud thins and its base rises, so that rain production decreases and more of the rain evaporates below cloud. In addition, w_e becomes more sensitive to droplet sedimentation (Fig. 7a). Together these changes shift the LWP maximum from $N_\mathrm{d} \geq 1000$ to about $100\,\mathrm{cm}^{-3}$. This picture is consistent with mixed-layer and LES arguments in which the turning point of the inverted-V relationship is set by the competition between precipitation and entrainment (Ackerman et al., 2004; Wood, 2007; Hoffmann et al., 2020, 2024). In EAMxx, the balance between these processes depends on horizontal grid spacing across the grey zone.

The extent to which this dependence reflects grey-zone physics rather than specific features of the configuration is not yet clear. Three model-specific factors co-vary with grid spacing in our experiment. SHOC's mixing length is capped at the grid spacing, and this cap is active throughout the boundary layer at 312 m. The physics time step and radiation interval decrease with grid spacing. The largest change in most quantities occurs between 10 and 5 km, where the resolved variance is still small and the circulations appear partly numerical. Any of these could contribute to the trends attributed above to grid spacing, and Sect. 4.4 lists the experiments that would separate them.

The weak negative branch at 312 m deserves particular caution. Its magnitude ($S \approx -0.04$ to -0.05) is an order of magnitude smaller than the steady-state adjustments inferred from LES of non-precipitating stratocumulus (Glassmeier et al., 2021) and smaller than the observed slopes, which are themselves upper bounds (Gryspeerdt et al., 2019). Possible reasons include the transience of the runs, the absence of a droplet-size dependence of cloud-top evaporation in SHOC's saturation adjustment (Wang et al., 2003; Hoffmann and Feingold, 2019), the 50 m vertical spacing at the inversion, and the humidity of the overlying air, which controls the strength of entrainment drying (Ackerman et al., 2004; Chen et al., 2014); in our runs the q_t jump at the inversion is 5.8–6.5 $\mathrm{g\,kg}^{-1}$.

4.2 Relation to other models

The closest prior study is that of Terai et al. (2020), who compared an ultra-parameterised multiscale model with 250 m cloud-resolving grids against a conventional multiscale model with 4 km grids. They found a weaker LWP increase with aerosol at the finer resolution, caused by weaker LWP increases in non-raining clouds and a smaller raining fraction. Our results show the same tendency for a single case in a doubly periodic configuration, in that the raining fraction and the precipitation-suppression term decrease with grid spacing, and LWP increases with N_d only weakly or not at all in the non-raining runs at sub-kilometre spacing. Sato et al. (2018) attributed the LWP decrease in their global cloud-system-resolving model to explicitly simulated condensation and evaporation responses to aerosol. That pathway is absent from EAMxx, in which condensation is diagnosed by SHOC's saturation adjustment, so the mechanisms in the two models differ.

The comparison with Mülmenstädt et al. (2024a) is instructive. Using E3SM single-column physics, they found that switching sedimentation off removed the LWP decrease and that doubling it made the response more negative, and they concluded that GCM-type physics can represent the sedimentation–entrainment coupling. In our coarse runs, which use SHOC rather than the turbulence scheme of the E3SM GCM, a 50 m vertical grid and a 100 s time step, entrainment is nearly insensitive to sedimentation, while the susceptibility with precipitation suppression on is sensitive to it through an interaction with autoconversion. The two results are not in conflict, because the configurations differ, but they show that the sedimentation–entrainment coupling in single-column physics depends on the turbulence scheme and its numerical settings. Our coarse runs are therefore not representative of GCM columns, which use much longer time steps, coarser vertical grids and different turbulence and microphysics schemes, and we do not draw conclusions about GCM behaviour from them. Likewise, GSRMs with other turbulence closures, such as Smagorinsky-type or scale-aware schemes that blend boundary-layer and LES closures (Boutle et al., 2014; Efstathiou and Beare, 2015), may behave differently in the grey zone. Within EAMxx, the susceptibility at 2.5–5 km (0.11–0.20 on the moderate branch, about 40–75 % of the 40 km value) lies between the values at 40 km and 312 m. These results suggest that estimates of aerosol–cloud adjustments from GSRMs should be accompanied by resolution-sensitivity tests, and they motivate grey-zone-aware turbulence closures. The factorial approach also offers process-level targets, the precipitation-suppression term and the sensitivity to sedimentation, that could help to reduce equifinality in constraining aerosol–cloud parameterisations (Stanford et al., 2026).

4.3 Relation to satellite-based estimates

Using model-diagnosed cloud-top N_d instead of layer-mean N_d changes S by at most 0.1 in these simulations, but this does not test retrieval biases, and satellite analyses also use in-cloud LWP at pixel scale rather than domain means. The within-scene covariation of LWP and cloud-top N_d with uniform aerosol spans a range of slopes much wider than the susceptibilities and changes sign with grid spacing. This adds a model-based illustration to the concerns raised by Ghan et al. (2016), Gryspeerdt et al. (2019), Arola et al. (2022) and Mahfouz et al. (2024) that spatial co-variability is not a measure of the adjustment, and it cautions against evaluating the LWP adjustments of kilometre-scale models with satellite statistics that do not account for cloud-controlled N_d variability.

4.4 Limitations and required experiments

The study simulates a single nocturnal, drizzling case with constant forcing and no shortwave radiation. The diurnal cycle modulates adjustments (Sandu et al., 2008), and the humidity above the inversion controls the sign of the entrainment response (Ackerman et al., 2004; Chun et al., 2023). Each configuration was run once. The quoted spread in S reflects temporal sampling only, and the small negative susceptibilities at sub-kilometre spacing (-0.02 to -0.09) require confirmation with initial-condition ensembles of three to five members, for example at target N_d of 100, 200 and 1000 cm^{-3} at 2.5 km and 312 m. After 12 h the sub-kilometre runs are still thinning by 4–6 % h^{-1}, and their susceptibilities have not levelled off, so a subset should be extended to 24 h or longer (Glassmeier et al., 2021; Bretherton et al., 2010).

The surface fluxes were prescribed. A bulk estimate that scales the latent heat flux with $q_{\mathrm{s}}(T_{\mathrm{s}}) - q_t$ and the sensible heat flux with $T_{\mathrm{s}} - T_{\mathrm{air}}$, calibrated to the prescribed fluxes and the initial mixed layer with a sea surface temperature of 292 K, suggests that interactive fluxes would have been 88–103 and 6–10 $\mathrm{W\,m^{-2}}$ at 312 m instead of 93 and 16 $\mathrm{W\,m^{-2}}$, and 77–79 and 17–20 $\mathrm{W\,m^{-2}}$ at $\Delta x \geq 10$ km. The feedback would thus reduce the warming of the sub-kilometre boundary layers and would slightly increase their moisture supply, but at 312 m the estimated latent heat flux remains below the entrainment drying of 105–131 $\mathrm{W\,m^{-2}}$. The estimate assumes fixed transfer coefficients and wind speed and should be tested with interactive surface fluxes over a fixed sea surface temperature. The absence of Coriolis force and wind nudging allowed the initial shear to decay, which may affect shear-driven entrainment in a resolution-dependent way.

The finest grid (312 m, with physics columns 469 m apart) is much coarser than the RF02 LES, and the vertical spacing at the inversion (about 50 m) was held fixed. Neither LWP nor $S_{100\to200}$ converged, so the results at 312 m describe the trend of a grey-zone model and not a converged answer. A converged reference with the same microphysics, for example an LES with KK2000 autoconversion at the same N_{d} levels, is not available. The physics time step and radiation interval co-varied with grid spacing, and SHOC's mixing length was capped at the grid spacing at 312 m. Experiments with a common time step at 40, 5 and 2.5 km, with a relaxed mixing-length cap at 625 and 312 m, and with refined vertical spacing near the inversion would separate these effects from grid spacing. The decomposition of the susceptibility depends on the matching N_{d} of the suppression-off configuration and on an extrapolation to zero sedimentation. Simulations with $s = 0$, with a second matching N_{d} (for example 200 or 1000 $\mathrm{cm^{-3}}$), or with a design in which the N_{d} seen by autoconversion and the N_{d} seen by sedimentation and radiation are varied independently, would give pivot-independent partial derivatives. The 240 km domain contains 11–19 cells at sub-kilometre spacing, and at 5 and 2.5 km the cell scale is probably limited by the effective resolution of the dynamical core.

CCN were prescribed and horizontally uniform, without aerosol processing, scavenging or replenishment, which can matter for drizzle-driven transitions (Kazil et al., 2011; Berner et al., 2013; Goren et al., 2019; Yamaguchi et al., 2017). Condensation and evaporation are computed by SHOC's saturation adjustment, so the droplet-size dependence of cloud-top evaporation that contributes to the entrainment pathway in LES (Wang et al., 2003; Bretherton et al., 2007; Hoffmann and Feingold, 2019) is absent. Autoconversion follows KK2000 without a subgrid variance enhancement, so at coarse spacing it uses the mean in-cloud cloud water, whereas at fine spacing resolved variability partly provides such an enhancement; other autoconversion schemes would change the precipitation-suppression term (Morrison et al., 2020). Surface precipitation includes the flux of cloud droplets reaching the surface, which we estimate to be negligible except in the coarse runs at target 20 $\mathrm{cm^{-3}}$ with $s = 2$ and suppression off. Finally, because the P3 sedimentation diagnostics were not usable, the sedimentation flux at cloud top was not diagnosed, and the role of sedimentation was inferred from the sedimentation-factor experiments.

5 Conclusions

We analysed 288 doubly periodic EAMxx simulations of the DYCOMS-II RF02 drizzling stratocumulus with horizontal grid spacings from 40 km to 312 m, target N_{d} from 20 to 1000 $\mathrm{cm^{-3}}$, three cloud-droplet sedimentation factors, and the N_{d} dependence of autoconversion switched on or off. Our conclusions are as follows.

1. The LWP response to N_d depends strongly on grid spacing. At 10–40 km LWP increased monotonically with N_d ($S_{100\to200} = 0.24$–0.27). With decreasing grid spacing the LWP maximum moved to lower N_d, reaching about 100 cm^{-3} at 312 m, where $S_{100\to200} = -0.05$ and $S_{200\to1000} = -0.04$ in single, still-evolving realisations.

2. At coarse spacing, removing the N_d dependence of autoconversion removed the positive response, but the sensitivity to sedimentation interacted strongly with autoconversion, and the partition between the two depends on the matching N_d of the experiment design. With decreasing grid spacing, the precipitation-suppression contribution on the moderate branch decreased from 0.34–0.39 to 0.04, and the gain in LWP from precipitation suppression at a diagnosed N_d of 500 cm^{-3} decreased from 110 to 6 $\mathrm{g\,m}^{-2}$.

3. At sub-kilometre spacing the weak decrease of LWP at high N_d was also present without the N_d dependence of autoconversion and became steeper with faster sedimentation on the polluted branch, consistent with a sedimentation–entrainment feedback. Its sign is not established, given its small magnitude, the transience of the runs and the absence of ensembles.

4. The ratio of entrainment warming to radiative cooling increased from about 0.5 at coarse spacing to 1.2–1.3 at 312 m, entrainment drying exceeded the prescribed surface moisture supply, and the clouds thinned, lifted and produced less rain, most of which evaporated below cloud. A mixed-layer budget associates the general thinning with this increase in entrainment, but it has little skill for the N_d dependence at $\Delta x \leq 2.5$ km.

5. Neither LWP nor its N_d response converged, and LWP decreased by 21–40 % between 625 and 312 m at fixed diagnosed N_d. SHOC's mixing length was capped by the grid spacing at 312 m, the time step co-varied with grid spacing, and 3D turbulence had a negligible effect. The 625 m runs were closest to the RF02 observations at hours 4–6, but no grid spacing matched all observed quantities.

6. Susceptibilities computed with model-diagnosed cloud-top N_d agreed closely with those computed with layer N_d, whereas within-scene LWP–N_d slopes with uniform aerosol ranged from -7.9 to $+3.4$ and changed sign with grid spacing.

In EAMxx, the simulated LWP adjustment in drizzling stratocumulus thus depends on how the boundary-layer turbulence is partitioned between the grid and the turbulence scheme, as well as on the microphysics. Initial-condition ensembles, longer and diurnally forced simulations, interactive surface fluxes, tests of time step and mixing-length limits, refined vertical grids, a second matching N_d for the suppression-off configuration and additional cases are needed to determine how far these results generalise.

Appendix A: Skill of the mixed-layer budget

Table A1 lists the skill of the mixed-layer budget for the LWP tendency at each grid spacing, for the raw tendencies over all 2 h windows from 2 to 12 h and for the anomalies relative to the mean over the six N_d levels at fixed grid spacing, sedimentation

Table A1. Skill of the mixed-layer budget for $d\mathrm{LWP}/dt$ by grid spacing (3D turbulence off). r: correlation between budget and simulated tendency; RMSE and SD in $\mathrm{g\,m^{-2}\,h^{-1}}$. "Raw": all 2 h windows (180 per grid spacing). "Anomaly": deviations from the mean over the six N_d levels at fixed grid spacing, sedimentation factor, suppression setting and window, for all windows and for hours 10–12.

		raw		anomaly, all	anomaly, hours 10–12		
Δx	r	RMSE	SD	r	r	RMSE	SD
40 km	0.87	3.9	6.6	0.93	0.93	1.9	4.5
20 km	0.83	4.0	6.4	0.91	0.87	2.3	3.6
10 km	0.79	4.5	6.8	0.87	0.23	3.4	2.7
5 km	0.90	5.1	11.2	0.89	0.85	3.0	5.8
2.5 km	0.62	4.4	4.7	0.77	-0.02	1.4	0.9
1.25 km	0.38	4.8	4.9	0.57	0.08	1.4	0.9
625 m	0.07	3.6	1.8	0.02	0.10	1.8	0.9
312 m	0.44	2.7	2.2	0.51	0.57	0.8	0.8

factor, suppression setting and window. Figure A1 shows the anomalies. At 40, 20 and 5 km the budget reproduces both the tendencies and their N_d dependence. At $\Delta x \leq 2.5$ km the correlation for the anomalies in hours 10–12 is between -0.02 and 0.57, and the root-mean-square error is comparable to or larger than the standard deviation of the anomalies. The largest departures in the raw tendencies occur where the boundary layer is decoupled and cloud base departs from the mixed-layer lifting condensation level. Using the simulated dz_b/dt instead of the estimate from the lifting condensation level degrades the agreement, because z_b is quantised by the vertical grid.

Appendix B: SHOC mixing length and activation floor

Figure B1a shows the domain-mean SHOC mixing length in hours 10–12 at target $200\,\mathrm{cm^{-3}}$. At $\Delta x \geq 10$ km it reaches about 1600 m in the upper boundary layer. At 5 km to 1.25 km it peaks at 900–1100 m near 0.85 z_i and is about 600–800 m below. At 625 m it is 480–610 m, close to the cap of 625 m, and at 312 m it equals the cap of 312 m nearly everywhere in the boundary layer. Figure B1b shows the in-cloud N_d profiles and the activation floor for target $200\,\mathrm{cm^{-3}}$. The in-cloud N_d follows the floor, which decreases with height because the prescribed CCN number does, so the diagnosed in-cloud N_d depends on the height of the cloud layer.

Appendix C: 3D turbulence

Figure C1 shows the difference between runs with 3D turbulence on and off for the 36 pairs at 625 m and the 35 completed pairs at 312 m. LWP changes by between -1.0 and $+1.4\,\mathrm{g\,m^{-2}}$, cloud cover by between -0.005 and $+0.007$, and the entrainment

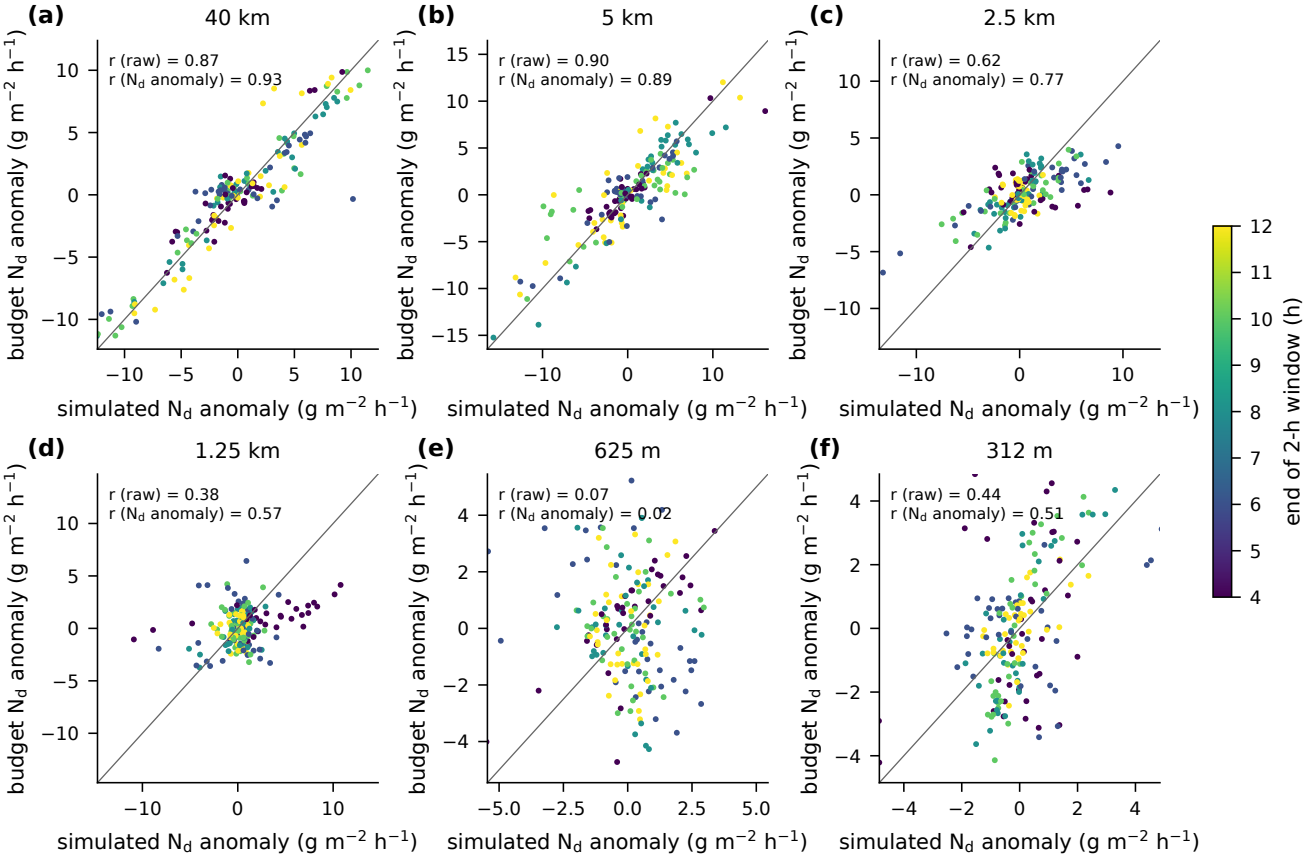


Figure A1. Anomalies of $d\mathrm{LWP}/dt$ relative to the mean over the six N_d levels, from the mixed-layer budget against the simulated values, for all 2 h windows (colour: end of window) at (a) 40 km, (b) 5 km, (c) 2.5 km, (d) 1.25 km, (e) 625 m and (f) 312 m. Each panel gives the correlation for the raw tendencies and for the anomalies.

rate by between -0.13 and $+0.02\,\mathrm{mm\,s^{-1}}$. The only systematic signal is a slight reduction of entrainment. These differences are an order of magnitude smaller than those between successive grid spacings or between N_d levels.

Appendix D: LWP spectra and distributions

Figure D1 shows the LWP spectra and column-LWP distributions referred to in Sect. 3.8. The spectral peak shifts from 48 km at 5 km to about 30 km at 2.5 km and to 13–22 km at 625 and 312 m. The LWP distributions broaden and shift to lower values as the grid is refined, and at target $N_\mathrm{d} = 20\,\mathrm{cm^{-3}}$ they develop a large population of clear or nearly clear columns.

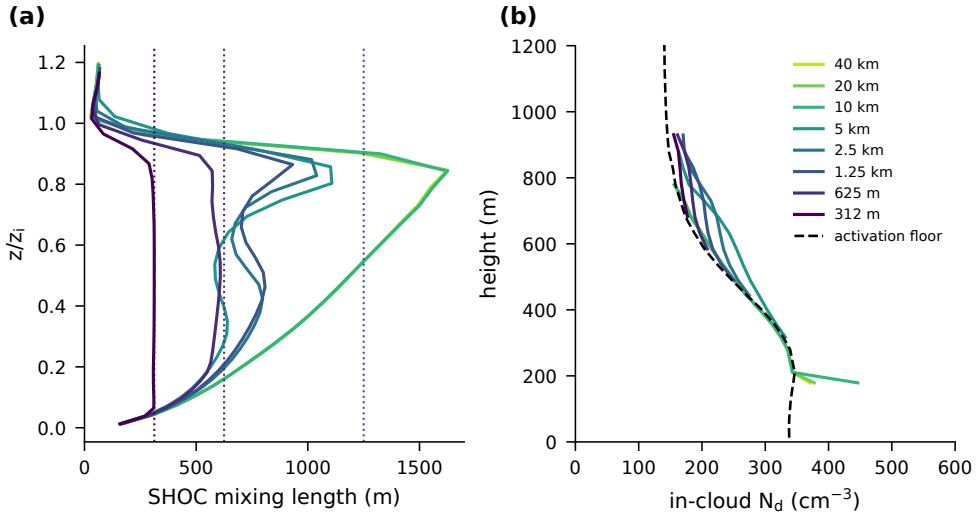


Figure B1. (a) Domain-mean SHOC mixing length against z/z_{i} and (b) in-cloud N_{d} (where the domain-mean liquid cloud fraction exceeds 0.2) with the activation floor (dashed), for target $N_{\mathrm{d}} = 200\,\mathrm{cm}^{-3}$, hours 10–12, suppression on and sedimentation factor 1. Dotted lines in (a) mark the grid-spacing cap at 1.25 km, 625 m and 312 m.

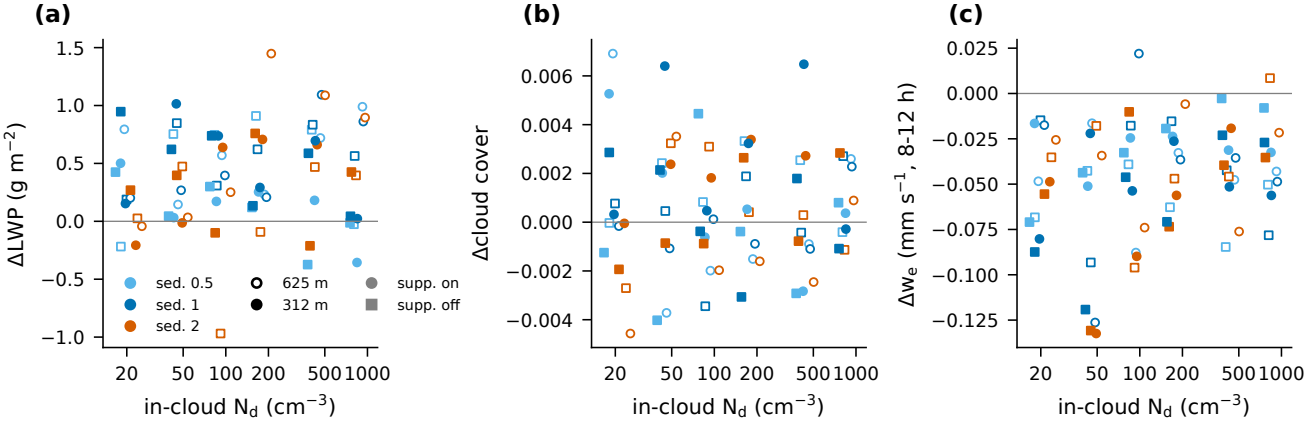


Figure C1. Difference (3D turbulence on minus off) in (a) LWP, (b) cloud cover (LWP above $10\,\mathrm{g\,m}^{-2}$) and (c) entrainment rate at 625 m (open symbols) and 312 m (filled symbols), coloured by sedimentation factor; circles: suppression on, squares: off.

Code and data availability. The model code is the E3SM branch `mahf708/eamxx/p3-cloud-sed-factor` at https://github.com/mahf708/E3SM/tree/mahf708/eamxx/p3-cloud-sed-factor, which contains E3SM pull request 9017 (the cloud-top output operator) and the P3 sedimentation-factor option. Run scripts and all analysis scripts used to produce the tables and figures are in the `sed-precsupp-study` directory of the branch https://github.com/mahf708/scmlib/tree/sed-precsupp-study. Model output and the derived tables are available from the corresponding contact on request.

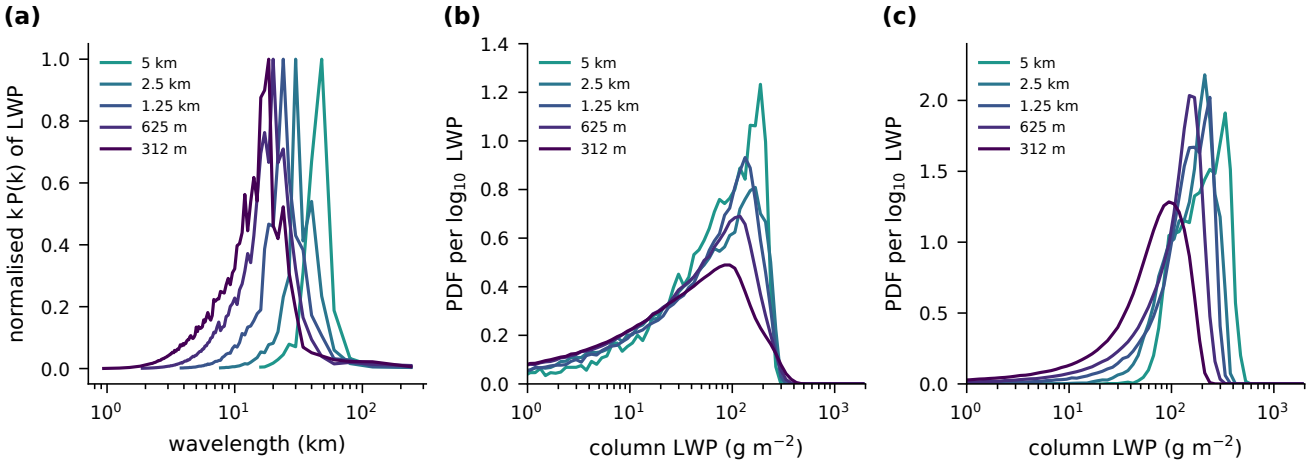


Figure D1. (a) Normalised radially averaged LWP spectra ($k\,P(k)$, hours 10–12) for target $N_\mathrm{d} = 200\,\mathrm{cm}^{-3}$. (b, c) Distributions of column LWP over hours 10–12 for target N_d of 20 and 1000 cm^{-3}. Suppression on, sedimentation factor 1. In (b), clear columns at small LWP are off scale.

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